

Zeta-Law Entropy, Gamma Curvature, and Laplace-Measure Applications

Pudim AI Project

Version v014 – 2026-06-04

Abstract

This paper develops a unified public theory built around Laplace-transform positivity. Zeta-law entropy is organized through logarithmic compression, gamma-ratio and polygamma inequalities are expressed by completely monotone kernels, and applications are obtained by transporting measure positivity through Stieltjes, Bernstein, stable-subordination, determinant, triangular, inverse-Laplace, derivative-chain, ratio-kernel, and moment-ratio mechanisms. The emphasis is on reusable transformations: source problems enter as fixed applications, while the proofs expose the structural tools that connect them.

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Keywords. Riemann zeta function; zeta probability law; entropy; Laplace transform; complete monotonicity; Bernstein functions; Stieltjes functions; gamma function; digamma function; Mills ratio; generalized Stieltjes functions.

1 Motivation and main results

Many classical identities for $\zeta(s)$ become more transparent after a probabilistic normalization. For $\beta > 1$, the weights $n^{-\beta}$ define a Gibbs law whose partition function is precisely $\zeta(\beta)$. The energy is $\log n$, the free energy is $\log \zeta(\beta)$, and ratios such as $\zeta(\beta+r)/\zeta(\beta)$ become moments. This viewpoint is elementary, but it packages several unrelated-looking zeta inequalities into a common calculus.

The same principle guides the special-function part of the paper. Complete monotonicity, Bernstein properties, Stieltjes representations, and endpoint obstructions are treated as transport questions: once a positive kernel, measure representation, determinant reduction, or rational certificate is identified, it can be reused across multiple source problems. The central theme is therefore not an individual inequality, but the mechanism that makes families of inequalities stable under analytic transformations.

Section 3 records solved external problems with stable application labels. The ledger is not a chronology; it is a public index of source questions and the results that resolve them. The body of the paper organizes those results by mechanism: zeta-law compression, gamma and polygamma curvature, Laplace and Stieltjes transport, determinant and moment-ratio reductions, endpoint obstructions, and explicit certificates. This organization lets the reader trace each application back to the tools that make it work.

2 Global notation and positivity conventions

The Riemann zeta function is denoted by ζ , and Euler’s gamma function is denoted by Γ . The digamma function is $\psi = \Gamma'/\Gamma$, its higher derivatives are the polygamma functions $\psi^{(m)}$, and Euler’s constant is $\gamma = -\psi(1)$. We write $\mathbb{N} = \{1, 2, \dots\}$, $\mathbb{N}_0 = \{0, 1, 2, \dots\}$, and use $\mathbb{E}_\mu$, $\mathbb{P}_\mu$, and Var_μ for expectation, probability, and variance under a measure μ . When the subscript is β , these symbols

refer to the zeta law $\rho_\beta(n) = n^{-\beta}/\zeta(\beta)$ on $\mathbb{N}$, later recorded formally in Definition 4.1. The rising factorial, or Pochhammer symbol, is $(a)_m = a(a+1)\cdots(a+m-1)$ for $m \geq 1$, with $(a)_0 = 1$. The notation $[z^m]F(z)$ means coefficient extraction from the power series F .

A C^∞ function $f : (0, \infty) \rightarrow \mathbb{R}$ is completely monotone if

$$(-1)^m f^{(m)}(x) \geq 0 \quad (m \in \mathbb{N}_0, x > 0),$$

and it is strictly completely monotone if all these inequalities are strict whenever the derivative is not identically zero. A positive function f is logarithmically completely monotone in this paper if $\log f$ has the same alternating derivative sign pattern on $(0, \infty)$. A nonnegative C^∞ function B is a Bernstein function if B' is completely monotone. A complete Bernstein function is a Bernstein function with the standard Stieltjes-density representation; equivalently, in the applications below, it is obtained by integrating a completely monotone density against $1 - e^{-xt}$. A Stieltjes function is a function representable in the form

$$f(x) = a + \frac{b}{x} + \int_0^\infty \frac{1}{x+t} d\mu(t), \quad a, b \geq 0,$$

with μ a positive measure satisfying the usual local integrability condition. A generalized Stieltjes function of order $\lambda > 0$ is a positive superposition of kernels $(x+t)^{-\lambda}$, in the sense used by Sokal. Source-specific special functions and measures are introduced locally before the theorem that uses them.

3 Applications

The application ledger is append-only public memory. Each row records the immutable APP identifier, the canonical short description, the result reference used by the paper, and the cited source problem.

1. **APP-0001.** Alzer-Kwong convexity and concavity problem. Solved by Theorem 9.2; source problem reference [30].
2. **APP-0002.** Nantomah zeta positivity problem. Solved by Theorem 9.1; source problem reference [1].
3. **APP-0003.** Sroysang generalized Holder problem. Solved by Theorem 9.3; source problem reference [2].
4. **APP-0004.** Complete monotonicity of $(\log \Gamma(x) + \log \Gamma(1/x))''$. Solved by Theorem 8.5; source problem reference [31].
5. **APP-0005.** Exact inverse-tail floor formula at $s=7$. Solved by Theorem 5.2; source problem reference [6].
6. **APP-0006.** Exact inverse-tail floor formula at $s=8$. Solved by Theorem 5.3; source problem reference [6].
7. **APP-0007.** Concavity or complete monotonicity of the polygamma product P0. Solved by Theorem 4.4; source problem reference [31].
8. **APP-0008.** Higher-order monotonicity of polygamma products Pn. Solved by Theorem 4.5; source problem reference [31].

9. **APP-0009.** Sharp reciprocal Gamma-product monotonicity threshold. Solved by Theorem 8.4; source problem reference [32].
10. **APP-0010.** Nielsen k -beta derivative-ratio monotonicity. Solved by Theorem 10.2; source problem reference [8].
11. **APP-0011.** Exact $n = 2$ beta-window endpoint. Solved by Theorem 11.1; source problem reference [31].
12. **APP-0012.** Baricz V_q strict q -log-convexity. Solved by Theorem 11.2; source problem reference [9].
13. **APP-0013.** Yin (p, k) -digamma sharp α -necessity. Solved by Theorem 11.3; source problem reference [10].
14. **APP-0014.** Ramanujan integral is Stieltjes and has a complete Bernstein primitive. Solved by Theorem 12.1; source problem reference [13].
15. **APP-0015.** Sibisi-Prabhakar measure is a stable-subordination push-forward. Solved by Theorem 12.2; source problem reference [14].
16. **APP-0016.** From's Mills-ratio bound extends to every derivative level. Solved by Theorem 12.3; source problem reference [15].
17. **APP-0017.** Baricz gamma-quotient Bernstein-for-all problem has a rational counterexample. Solved by Theorem 12.4; source problem reference [16].
18. **APP-0018.** Qi-Guo tau threshold has a supremum-corrected exact value. Solved by Theorem 12.5; source problem reference [17].
19. **APP-0019.** Wakrim W-symbol Bernstein range closes the beta gap. Solved by Theorem 12.6; source problem reference [18].
20. **APP-0020.** Qi's degree-four conjecture for h_λ is refuted at the endpoint. Solved by Theorem 12.7; source problem reference [19].
21. **APP-0021.** Szabo's cutoff problem has $\alpha_0 = 1$. Solved by Theorem 12.8; source problem reference [20].
22. **APP-0022.** Chen-Choi Gurland gamma-ratio conjecture is logarithmically completely monotone. Solved by Theorem 12.9; source problem reference [21].
23. **APP-0023.** Sokal generalized-Stieltjes lambda-derivatives have a triangular nonnegative compression. Solved by Theorem 12.10; source problem reference [22].
24. **APP-0024.** Simon gamma quotient is Bernstein. Solved by Theorem 12.11; source problem reference [23].
25. **APP-0025.** Du-Wang h_3 monotonicity is classified. Solved by Theorem 12.12; source problem reference [24].
26. **APP-0026.** Baskakov even-power complete-monotonicity conjecture is false. Solved by Theorem 12.13; source problem reference [25].

27. **APP-0027.** Ma-Weigert derivative regions form a descending chain. Solved by Theorem [12.14](#); source problem reference [\[11\]](#).
28. **APP-0028.** Qi-Agarwal/Yin divisor-polygamma parity is corrected. Solved by Theorem [12.15](#); source problem reference [\[12\]](#).
29. **APP-0029.** Bulboaca-Zayed Gamma quotient has one critical point. Solved by Theorem [12.16](#); source problem reference [\[5\]](#).
30. **APP-0030.** Ramanujan integral Turan window is false. Solved by Theorem [12.17](#); source problem reference [\[13\]](#).
31. **APP-0031.** Baricz gamma-quotient Bernstein test is false. Solved by application ledger entry [31](#); source problem reference [\[16\]](#).
32. **APP-0032.** From's all- L Mills-ratio bound family is explicit. Solved by application ledger entry [32](#); source problem reference [\[15\]](#).
33. **APP-0033.** Ramanujan antiderivative is complete Bernstein. Solved by application ledger entry [33](#); source problem reference [\[13\]](#).
34. **APP-0034.** Bulboaca-Zayed gamma-quotient monotonicity solved. Solved by application ledger entry [34](#); source problem reference [\[5\]](#).
35. **APP-0035.** Keady self-bijection inverse-CM question is negative. Solved by application ledger entry [35](#); source problem reference [\[33\]](#).
36. **APP-0036.** Baskakov even-line complete-monotonicity fails at $r = 8$. Solved by application ledger entry [36](#); source problem reference [\[25\]](#).
37. **APP-0037.** Du-Wang h_3 is increasing exactly on $\frac{1}{2} \leq a \leq 1$. Solved by application ledger entry [37](#); source problem reference [\[24\]](#).
38. **APP-0038.** Ma-Weigert derivative-sign regions form a chain. Solved by application ledger entry [38](#); source problem reference [\[11\]](#).
39. **APP-0039.** Ramanujan Turan-window complete-monotonicity interval is false. Solved by Counterexample [13.1](#); source problem reference [\[13\]](#).
40. **APP-0040.** Yang-Tian Bessel- W power-Bernstein conjecture is true. Solved by Corollary [13.3](#); source problem reference [\[34\]](#).
41. **APP-0041.** Qi h_λ degree-four complete-monotonicity conjecture is false. Solved by Counterexample [13.4](#); source problem reference [\[19\]](#).
42. **APP-0042.** Modified Bessel square-root log-concavity fails. Solved by Counterexample [14.1](#); source problem reference [\[35\]](#).
43. **APP-0043.** Riccati log-concavity inequality for I_ν is false. Solved by Corollary [14.2](#); source problem reference [\[35\]](#).
44. **APP-0044.** Bessel ratio quadratic lower bound is false. Solved by Counterexample [14.3](#); source problem reference [\[35\]](#).
45. **APP-0045.** Three-regime Bessel log-concavity certificate route is refuted. Solved by Corollary [14.4](#); source problem reference [\[35\]](#).

4 Zeta-law calculus

Definition 4.1 (Riemann zeta probability law). For $\beta > 1$, define a probability law ρ_β on $\mathbb{N}$ by

$$\rho_\beta(n) = \frac{n^{-\beta}}{\zeta(\beta)}, \quad n \in \mathbb{N}. \quad (4.1)$$

Equivalently, with $E(n) = \log n$ and $Z(\beta) = \zeta(\beta)$,

$$\rho_\beta(n) = \frac{e^{-\beta E(n)}}{Z(\beta)}. \quad (4.2)$$

Definition 4.2 (Zeta free energy). The zeta free energy is

$$A(\beta) = \log \zeta(\beta), \quad \beta > 1. \quad (4.3)$$

Proposition 4.3 (Free-energy derivatives). For every $\beta > 1$,

$$A'(\beta) = -\mathbb{E}_\beta[\log N], \quad A''(\beta) = \text{Var}_\beta(\log N). \quad (4.4)$$

Proof. Absolute convergence of the differentiated Dirichlet series on compact subintervals of $(1, \infty)$ permits termwise differentiation. Thus

$$A'(\beta) = \frac{\zeta'(\beta)}{\zeta(\beta)} = -\frac{\sum_{n \geq 1} (\log n) n^{-\beta}}{\zeta(\beta)} = -\mathbb{E}_\beta[\log N]. \quad (4.5)$$

Differentiating once more gives

$$A''(\beta) = \mathbb{E}_\beta[(\log N)^2] - \mathbb{E}_\beta[\log N]^2 = \text{Var}_\beta(\log N). \quad (4.6)$$

□

Qi, Lim, and Nantomah ask whether $P_0(x) = \psi(x)\psi(1/x)$ is concave, and more strongly whether $-P_0''$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers the stronger question.

Theorem 4.4 (APP-0007: Complete monotonicity of reciprocal digamma product curvature). *Let*

$$P_0(x) = \psi(x)\psi(1/x), \quad x > 0. \quad (4.7)$$

Then $-P_0''$ is strictly completely monotonic on $(0, \infty)$. In particular, P_0 is strictly concave on $(0, \infty)$.

Proof. Direct differentiation gives

$$\begin{aligned} P_0''(x) &= \psi''(x)\psi(1/x) - \frac{2}{x^2}\psi'(x)\psi'(1/x) \\ &\quad + \frac{2}{x^3}\psi(x)\psi'(1/x) + \frac{1}{x^4}\psi(x)\psi''(1/x). \end{aligned} \quad (4.8)$$

Use the shifted Weierstrass expansion

$$\psi(x) = -\gamma - \frac{1}{x} + \sum_{m=1}^{\infty} a_m(x), \quad a_m(x) = \frac{x}{m(m+x)}, \quad (4.9)$$

and, after substituting $1/x$,

$$\psi(1/x) = -\gamma - x + \sum_{n=1}^{\infty} b_n(x), \quad b_n(x) = \frac{1}{n(nx+1)}. \quad (4.10)$$

On compact subintervals of $(0, \infty)$, the series for a_m , b_n , and their first two derivatives are uniformly summable; the differentiated double products are dominated by products of summable majorants. Hence P_0 may be differentiated term by term twice.

Let $A_0(x) = -\gamma - 1/x$ and $B_0(x) = -\gamma - x$. The base term satisfies

$$-\frac{d^2}{dx^2}(A_0(x)B_0(x)) = -\frac{2\gamma}{x^3}. \quad (4.11)$$

For $n \geq 1$, elementary rational differentiation and Laplace inversion give

$$-\frac{d^2}{dx^2}(A_0(x)b_n(x)) = \int_0^{\infty} e^{-xt} t^2 \frac{n + (\gamma - n)e^{-t/n}}{n^2} dt, \quad (4.12)$$

and, for $m \geq 1$,

$$-\frac{d^2}{dx^2}(B_0(x)a_m(x)) = \int_0^{\infty} e^{-xt} t^2 (m - \gamma) e^{-mt} dt. \quad (4.13)$$

For $mn > 1$,

$$-\frac{d^2}{dx^2}(a_m(x)b_n(x)) = \int_0^{\infty} e^{-xt} t^2 \left[\frac{e^{-t/n}}{mn^2(mn-1)} - \frac{e^{-mt}}{n(mn-1)} \right] dt. \quad (4.14)$$

The exceptional product is $a_1(x)b_1(x) = x/(x+1)^2$, and

$$-\frac{d^2}{dx^2} \frac{x}{(x+1)^2} = \int_0^{\infty} e^{-xt} t^2 (t-1) e^{-t} dt. \quad (4.15)$$

The identities above follow from

$$\frac{1}{(x+c)^3} = \int_0^{\infty} e^{-xt} \frac{t^2}{2} e^{-ct} dt, \quad \frac{1}{(x+c)^4} = \int_0^{\infty} e^{-xt} \frac{t^3}{6} e^{-ct} dt. \quad (4.16)$$

The coefficients of the double products group by the two identities

$$\sum_{m=1}^{\infty} \frac{1}{m(mn-1)} = -\psi(1-1/n) - \gamma \quad (n \geq 2) \quad (4.17)$$

and

$$\sum_{n=1}^{\infty} \frac{1}{n(mn-1)} = -\psi(1-1/m) - \gamma \quad (m \geq 2). \quad (4.18)$$

Both are immediate from $\psi(z+1) + \gamma = \sum_{k=1}^{\infty} z/(k(k+z))$, with $z = -1/n$ and $z = -1/m$. Combining (4.11)–(4.18) gives

$$-P_0''(x) = \int_0^{\infty} e^{-xt} K(t) dt, \quad (4.19)$$

where

$$K(t) = t^2 \left[1 - \gamma + (t-1)e^{-t} + \sum_{n=2}^{\infty} \frac{n(1 - e^{-t/n}) - \psi(1 - 1/n)e^{-t/n}}{n^2} + \sum_{m=2}^{\infty} (m + \psi(1 - 1/m))e^{-mt} \right]. \quad (4.20)$$

It remains to prove $K(t) > 0$. Since ψ is increasing and $1 - 1/n \in [1/2, 1)$,

$$-\psi(1 - 1/n) > 0 \quad (n \geq 2). \quad (4.21)$$

Also

$$m + \psi(1 - 1/m) \geq 2 - \gamma - 2 \log 2 > 0 \quad (m \geq 2), \quad (4.22)$$

because $\psi(1 - 1/m) \geq \psi(1/2) = -\gamma - 2 \log 2$. Thus the two sums in (4.20) are positive term by term, apart from the possible need to dominate $1 - \gamma + (t-1)e^{-t}$.

For $n \geq 2$,

$$-\psi(1 - 1/n) = \gamma + \int_{1-1/n}^1 \psi'(u) du > \gamma + \int_{1-1/n}^1 \frac{du}{u^2} = \gamma + \frac{1}{n-1}. \quad (4.23)$$

Consequently the n -sum in (4.20) is bounded below by

$$Ae^{-t/2}, \quad A = \gamma(\zeta(2) - 1) + 2 - \zeta(2). \quad (4.24)$$

Therefore $K(t)/t^2$ is bounded below by

$$f(t) = 1 - \gamma + (t-1)e^{-t} + Ae^{-t/2}. \quad (4.25)$$

Now

$$f(0) = A - \gamma = (2 - \zeta(2))(1 - \gamma) > 0, \quad \lim_{t \rightarrow \infty} f(t) = 1 - \gamma > 0. \quad (4.26)$$

Moreover,

$$f'(t) = e^{-t} \left(2 - t - \frac{A}{2} e^{t/2} \right), \quad (4.27)$$

and the factor in parentheses is strictly decreasing. Hence f has at most one critical point, and any such point is a maximum. Its minimum on $[0, \infty)$ is attained at an endpoint, so $f(t) > 0$. Thus $K(t) > 0$ for $t > 0$.

Differentiating under (4.19) yields

$$(-1)^r \frac{d^r}{dx^r} (-P_0''(x)) = \int_0^\infty t^r e^{-xt} K(t) dt > 0 \quad (4.28)$$

for all $r \geq 0$. This proves strict complete monotonicity of $-P_0''$, and the case $r = 0$ gives strict concavity of P_0 . $\square$

Qi, Lim, and Nantomah also ask whether the higher-order products $P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x)$, $n \in \mathbb{N}$, are convex, and more strongly whether P_n'' is completely monotonic. The following result refutes the stronger assertion. It does not settle the weaker convexity question.

Theorem 4.5 (APP-0008: Counterexample to complete monotonicity of higher-order polygamma product curvature). *Let*

$$P_n(x) = \psi^{(n)}(x)\psi^{(n)}(1/x), \quad n \in \mathbb{N}, \quad x > 0. \quad (4.29)$$

Then P_n'' is not completely monotonic for all $n \in \mathbb{N}$. More precisely, for every $n \geq 29$,

$$P_n'''(2) > 0, \quad (4.30)$$

and, already at lower order,

$$P_7^{(6)}(3) < 0. \quad (4.31)$$

Proof. For $m \geq 1$, define the sign-normalized polygamma function

$$A_m(x) = (-1)^{m+1}\psi^{(m)}(x) = m! \sum_{k=0}^{\infty} (x+k)^{-m-1}. \quad (4.32)$$

Then $A_m(x) > 0$, $A'_m(x) = -A_{m+1}(x)$, and $P_n(x) = A_n(x)A_n(1/x)$. Differentiating twice gives

$$\begin{aligned} P_n''(x) = & A_{n+2}(x)A_n(1/x) - \frac{2}{x^2}A_{n+1}(x)A_{n+1}(1/x) \\ & - \frac{2}{x^3}A_n(x)A_{n+1}(1/x) + \frac{1}{x^4}A_n(x)A_{n+2}(1/x). \end{aligned} \quad (4.33)$$

First prove (4.30). Put $p = n + 1$, and normalize

$$Q_p(x) = \frac{P_n(x)}{(n!)^2} = \sum_{m,\ell \geq 0} \left(\frac{x}{(x+m)(1+\ell x)} \right)^p. \quad (4.34)$$

At $x = 2$, the term $(m, \ell) = (0, 0)$ is constant, and the unique largest nonconstant base is $(m, \ell) = (1, 0)$, equal to $2/3$. Write

$$Q_p(x) = 1 + \left(\frac{x}{x+1} \right)^p + R_p(x). \quad (4.35)$$

For $h(x) = x/(x+1)$, direct differentiation gives

$$\frac{d^3}{dx^3} h(x)^p = p \left(\frac{x}{x+1} \right)^p \frac{p^2 - 6px - 3p + 6x^2 + 6x + 2}{x^3(x+1)^3}. \quad (4.36)$$

Therefore

$$\left. \frac{d^3}{dx^3} h(x)^p \right|_{x=2} = \frac{p(p^2 - 15p + 38)}{216} \left(\frac{2}{3} \right)^p. \quad (4.37)$$

For $p \geq 30$, this is at least

$$\frac{p^3}{432} \left(\frac{2}{3} \right)^p. \quad (4.38)$$

For a general summand

$$F_{m,\ell}(x) = \left(\frac{x}{(x+m)(1+\ell x)} \right)^p, \quad (4.39)$$

let $u_{m,\ell} = d(\log F_{m,\ell}^{1/p})/dx$. At $x = 2$,

$$|u_{m,\ell}| \leq \frac{1}{2}, \quad |u'_{m,\ell}| \leq \frac{1}{4}, \quad |u''_{m,\ell}| \leq \frac{1}{4}. \quad (4.40)$$

Since

$$F_{m,\ell}''' = F_{m,\ell} (p^3 u_{m,\ell}^3 + 3p^2 u_{m,\ell} u_{m,\ell}' + p u_{m,\ell}''), \quad (4.41)$$

we have $|F_{m,\ell}'''(2)| \leq p^3 F_{m,\ell}(2)$ for $p \geq 2$. The remaining mass in R_p is bounded by

$$\sum_{(m,\ell) \neq (0,0), (1,0)} F_{m,\ell}(2) < 8 \cdot 2^{-p} \quad (p \geq 3). \quad (4.42)$$

Indeed, the $m \geq 2, \ell = 0$ part is at most $3 \cdot 2^{-p}$, while the $\ell \geq 1$ part is at most $2 \cdot 2^{-p}(1 + (2/3)^p + 3 \cdot 2^{-p})$. Combining (4.38) and (4.42),

$$Q_p'''(2) \geq p^3 2^{-p} \left[\frac{1}{432} \left(\frac{4}{3} \right)^p - 8 \right]. \quad (4.43)$$

Because $(4/3)^{30} > 432 \cdot 8$, (4.43) is positive for $p \geq 30$. Hence $P_n'''(2) = (n!)^2 Q_{n+1}'''(2) > 0$ for $n \geq 29$. Complete monotonicity of P_n'' would require $(P_n'')' = P_n''' \leq 0$, so P_n'' is not completely monotonic for $n \geq 29$.

We now record the lower-order obstruction (4.31). For bookkeeping, put

$$T_{p,i,j}(x) = x^{-p} A_i(x) A_j(1/x). \quad (4.44)$$

The identity $A'_m = -A_{m+1}$ gives the exact recurrence

$$\frac{d}{dx} T_{p,i,j}(x) = -p T_{p+1,i,j}(x) - T_{p,i+1,j}(x) + T_{p+2,i,j+1}(x). \quad (4.45)$$

Starting with (4.33), applying (4.45) four times with $n = 7$ expresses $P_7^{(6)}(3)$ as an exact finite linear combination of 22 terms $T_{p,i,j}(3)$.

The sign of this finite expression can be certified by rational intervals. For rational $x > 0$ and $N \geq 1$, monotone integral comparison in (4.32) gives

$$m! \left(\sum_{k=0}^{N-1} (x+k)^{-m-1} + \frac{(x+N)^{-m}}{m} \right) \leq A_m(x), \quad (4.46)$$

and

$$A_m(x) \leq m! \left(\sum_{k=0}^{N-1} (x+k)^{-m-1} + \frac{(x+N)^{-m}}{m} + (x+N)^{-m-1} \right). \quad (4.47)$$

Using $N = 80$ in (4.46) and (4.47) for every $A_m(3)$ and $A_m(1/3)$ appearing in the 22-term expression, and choosing upper or lower endpoints according to the sign of each coefficient, yields

$$-74998.154649422399828421756528409981771664439555587 < P_7^{(6)}(3), \quad (4.48)$$

and

$$P_7^{(6)}(3) < -74997.331312439140215056900814096498575289999496021. \quad (4.49)$$

Thus $P_7^{(6)}(3) < 0$. If P_7'' were completely monotonic, then

$$(-1)^4 \frac{d^4}{dx^4} P_7''(3) = P_7^{(6)}(3) \geq 0, \quad (4.50)$$

contradicting (4.49). Hence the proposed complete-monotonicity strengthening for all $n \in \mathbb{N}$ is false in both an infinite high-order family and already at $n = 7$. $\square$

Proposition 4.6 (Zeta-law calculus). *Let $\beta > 1$, and let N have law ρ_β . For every $r \geq 0$,*

$$\mathbb{E}_\beta[N^{-r}] = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \quad (4.51)$$

Moreover, for $\alpha, \beta > 1$,

$$D(\rho_\alpha \| \rho_\beta) = (\beta - \alpha) \mathbb{E}_\alpha[\log N] + \log \zeta(\beta) - \log \zeta(\alpha). \quad (4.52)$$

Proof. The moment identity follows by direct summation:

$$\mathbb{E}_\beta[N^{-r}] = \sum_{n=1}^{\infty} n^{-r} \frac{n^{-\beta}}{\zeta(\beta)} = \frac{\zeta(\beta + r)}{\zeta(\beta)}. \quad (4.53)$$

For relative entropy, substitute (4.1):

$$D(\rho_\alpha \| \rho_\beta) = \sum_{n=1}^{\infty} \rho_\alpha(n) ((\beta - \alpha) \log n + \log \zeta(\beta) - \log \zeta(\alpha)), \quad (4.54)$$

which is (4.52). □

5 Tail zeta laws and inverse-tail floors

Definition 5.1 (Tail zeta partition function). For $s > 1$ and $n \in \mathbb{N}$, define the tail zeta partition function by

$$Z_{s,n} = \zeta_n(s) = \sum_{k=n}^{\infty} k^{-s}. \quad (5.1)$$

The corresponding tail zeta law is the probability distribution on $\{n, n+1, \dots\}$ given by

$$\rho_{s,n}(k) = \frac{k^{-s}}{\zeta_n(s)}, \quad k \geq n. \quad (5.2)$$

Exact formulas for integer parts of reciprocal zeta tails have been studied for small integer exponents. Known formulas include $s = 2, 3, 4, 5$, with a sufficiently-large- n formula for $s = 6$; Kim and Song record that no analogous formula was known for integer $s > 6$ at the time of their work [6]. The 2024 asymptotic inverse-tail results of Pan and Wu provide related context [7]. The next two theorems give exact formulas at $s = 7$ and $s = 8$. The first uses a single rational telescoping enclosure, while the second needs adjacent corrected asymptotic truncations.

Theorem 5.2 (APP-0005: Exact inverse-tail floor formula at $s=7$). *Let*

$$T_7(n) = \zeta_n(7) = \sum_{k=n}^{\infty} \frac{1}{k^7}. \quad (5.3)$$

Define

$$Q(n) = 120n^6 - 360n^5 + 660n^4 - 720n^3 + 354n^2 - 54n + 375, \quad P(n) = \frac{Q(n)}{20}. \quad (5.4)$$

Then, for every $n \geq 28$,

$$\lfloor T_7(n)^{-1} \rfloor = \lfloor P(n) \rfloor. \quad (5.5)$$

For $1 \leq n \leq 27$, the exact values are

n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$	n	$\lfloor T_7(n)^{-1} \rfloor$
1	0	10	4495760	19	241765529
2	119	11	8167814	20	331399044
3	1862	12	14061542	21	447175152
4	12573	13	23143975	22	594869693
5	54069	14	36668777	23	781167220
6	175597	15	56228085	24	1013751716
7	471118	16	83808666	25	1301401638
8	1100904	17	121852400	26	1654089273
9	2317459	18	173321080	27	2083084422

(5.6)

Proof. The polynomial Q is positive on $\mathbb{N}$, because

$$Q(n) = 120(n-1)^6 + 360(n-1)^5 + 660(n-1)^4 + 720(n-1)^3 + 354(n-1)^2 + 54(n-1) + 375. \quad (5.7)$$

For $k \geq 1$, direct expansion gives

$$\frac{1}{k^7} - \left(\frac{20}{Q(k)} - \frac{20}{Q(k+1)} \right) = \frac{9(60284k^4 + 29176k^2 + 15625)}{k^7 Q(k) Q(k+1)} > 0. \quad (5.8)$$

Summing (5.8) over $k \geq n$ yields

$$T_7(n) > \frac{20}{Q(n)} = \frac{1}{P(n)}, \quad (5.9)$$

and therefore $T_7(n)^{-1} < P(n)$.

For $k \geq 28$, another expansion gives

$$\left(\frac{20}{Q(k)-3} - \frac{20}{Q(k+1)-3} \right) - \frac{1}{k^7} = \frac{36(20k^6 - 14961k^4 - 7235k^2 - 3844)}{k^7(Q(k)-3)(Q(k+1)-3)}. \quad (5.10)$$

The numerator in (5.10) is positive for $k \geq 28$. Indeed, writing $k = m + 28$, it becomes

$$20m^6 + 3360m^5 + 220239m^4 + 7105168m^3 + 114013021m^2 + 751143512m + 436261580, \quad (5.11)$$

which has positive coefficients for $m \geq 0$. Hence, for $n \geq 28$,

$$T_7(n) < \frac{20}{Q(n)-3} = \frac{1}{P(n)-3/20}. \quad (5.12)$$

Combining (5.9) and (5.12) gives

$$P(n) - \frac{3}{20} < T_7(n)^{-1} < P(n). \quad (5.13)$$

Modulo 20,

$$Q(n) \equiv 14n^2 - 14n + 15 \pmod{20}. \quad (5.14)$$

Since $n(n-1) \pmod{10} \in \{0, 2, 6\}$, this implies

$$Q(n) \pmod{20} \in \{3, 15, 19\}. \quad (5.15)$$

Thus the fractional part of $P(n)$ is one of $3/20, 15/20, 19/20$. If $a = \lfloor P(n) \rfloor$, then $P(n) - 3/20 \geq a$, and (5.13) gives

$$a < T_7(n)^{-1} < a + 1. \quad (5.16)$$

This proves (5.5) for $n \geq 28$.

For the finite range, $T_7(1) > 1$, so $0 < T_7(1)^{-1} < 1$. For $2 \leq n \leq 27$, let M_n denote the corresponding value in (5.6). With $N = 1500$, exact rational arithmetic verifies

$$\sum_{k=n}^N \frac{1}{k^7} > \frac{1}{M_n + 1} \quad (5.17)$$

and

$$\sum_{k=n}^N \frac{1}{k^7} + \frac{1}{6N^6} < \frac{1}{M_n}. \quad (5.18)$$

The monotone integral estimate

$$\sum_{k=N+1}^{\infty} \frac{1}{k^7} < \int_N^{\infty} x^{-7} dx = \frac{1}{6N^6} \quad (5.19)$$

then gives

$$\frac{1}{M_n + 1} < T_7(n) < \frac{1}{M_n}. \quad (5.20)$$

Therefore $M_n < T_7(n)^{-1} < M_n + 1$, proving (5.6). The finite rational certificate is stored with the source as a reproducible calculation. $\square$

Theorem 5.3 (APP-0006: Exact inverse-tail floor formula at $s=8$). *Let*

$$T_8(n) = \zeta_n(8) = \sum_{k=n}^{\infty} \frac{1}{k^8}. \quad (5.21)$$

Define

$$\begin{aligned} A_8(n) = & 7n^7 - \frac{49}{2}n^6 + \frac{637}{12}n^5 - \frac{1715}{24}n^4 + \frac{7399}{144}n^3 - \frac{1715}{96}n^2 + \frac{69685}{1728}n - \frac{65611}{3456} \\ & - \frac{23764069}{103680n} - \frac{23764069}{207360n^2} + \frac{2102131409}{1244160n^3} + \frac{2149659547}{829440n^4} \\ & - \frac{203290800013}{14929920n^5} - \frac{1146003910541}{29859840n^6} + \frac{21587525169497}{179159040n^7}. \end{aligned} \quad (5.22)$$

Then, for every $n \geq 6$,

$$\lfloor T_8(n)^{-1} \rfloor = \lfloor A_8(n) \rfloor. \quad (5.23)$$

For $1 \leq n \leq 5$, the exact values are

$$\frac{n}{\lfloor T_8(n)^{-1} \rfloor} \left| \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \\ 0 & 245 & 5844 & 53503 & 291407 \end{array} \right|. \quad (5.24)$$

Proof. Set

$$B_8(n) = A_8(n) + \frac{200335112892761}{358318080n^8}. \quad (5.25)$$

For $C = A_8$ or B_8 , write $R_C(k) = 1/C(k)$. Exact denominator clearing gives the two telescoping inequalities

$$R_{A_8}(k) - R_{A_8}(k+1) - \frac{1}{k^8} > 0 \quad (k \geq 3) \quad (5.26)$$

and

$$\frac{1}{k^8} - (R_{B_8}(k) - R_{B_8}(k+1)) > 0 \quad (k \geq 1). \quad (5.27)$$

In (5.26), the cleared numerator is a polynomial in $k-3$ with all coefficients positive; in (5.27), the cleared numerator is a polynomial in $k-1$ with all coefficients positive. The exact coefficient verification is recorded in the rational certificate accompanying the manuscript.

Summing (5.26) over $k \geq n$ gives

$$T_8(n) < R_{A_8}(n) = \frac{1}{A_8(n)} \quad (5.28)$$

for $n \geq 3$, and hence $A_8(n) < T_8(n)^{-1}$. Summing (5.27) gives

$$T_8(n) > R_{B_8}(n) = \frac{1}{B_8(n)}, \quad (5.29)$$

and hence $T_8(n)^{-1} < B_8(n)$. Therefore, for $n \geq 6$,

$$A_8(n) < T_8(n)^{-1} < B_8(n). \quad (5.30)$$

It remains to see that no integer lies between $A_8(n)$ and $B_8(n)$. Exact integer arithmetic verifies

$$\lfloor A_8(n) \rfloor = \lfloor B_8(n) \rfloor \quad (6 \leq n \leq 10^6). \quad (5.31)$$

For $n > 10^6$, let $P_8(n)$ be the polynomial part of $A_8(n)$, through the constant term. The numerator of $P_8(n)$ modulo 3456 runs through the odd residue classes, so $P_8(n)$ is at distance at least $1/3456$ from the next integer. The correction $A_8(n) - P_8(n)$ is negative and has absolute value $< 1/3456$, while

$$0 < B_8(n) - A_8(n) < \frac{1}{3456}. \quad (5.32)$$

Thus $B_8(n)$ is still below the next integer after $A_8(n)$. Combining this with (5.30) proves (5.23).

For $1 \leq n \leq 5$, (5.24) is certified by exact rational partial sums. With $N = 300$, the displayed value M_n satisfies, for $2 \leq n \leq 5$,

$$\sum_{k=n}^N \frac{1}{k^8} > \frac{1}{M_n + 1} \quad (5.33)$$

and

$$\sum_{k=n}^N \frac{1}{k^8} + \frac{1}{7N^7} < \frac{1}{M_n}. \quad (5.34)$$

Since

$$\sum_{k=N+1}^{\infty} \frac{1}{k^8} < \int_N^{\infty} x^{-8} dx = \frac{1}{7N^7}, \quad (5.35)$$

we get $M_n < T_8(n)^{-1} < M_n + 1$. For $n = 1$, $T_8(1) > 1$, so the floor is 0. $\square$

6 Modular successor entropy

Definition 6.1 (Modular residue distribution and successor entropy). For $q \geq 2$, define the residue distribution of ρ_β modulo q by

$$\mu_{q,\beta}(a) = \sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n), \quad a \in \mathbb{Z}/q\mathbb{Z}. \quad (6.1)$$

Writing $a + 1$ cyclically in $\mathbb{Z}/q\mathbb{Z}$, define

$$B_q(\beta) = \sum_{a \in \mathbb{Z}/q\mathbb{Z}} \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\mu_{q,\beta}(a+1)}. \quad (6.2)$$

Theorem 6.2 (Zeta-law successor entropy and modular resolution). *For every $\beta > 1$, define*

$$\Sigma(\beta) = \sum_{n=1}^{\infty} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (6.3)$$

Then

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \zeta(\beta+k), \quad (6.4)$$

and

$$\Sigma(\beta) = \sup_{q \geq 2} B_q(\beta) = \lim_{q \rightarrow \infty} B_q(\beta). \quad (6.5)$$

Proof. Since

$$\frac{\rho_\beta(n)}{\rho_\beta(n+1)} = \left(1 + \frac{1}{n}\right)^\beta, \quad (6.6)$$

we have

$$\Sigma(\beta) = \frac{\beta}{\zeta(\beta)} \sum_{n=1}^{\infty} n^{-\beta} \log \left(1 + \frac{1}{n}\right). \quad (6.7)$$

Using

$$\log(1+x) = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} x^k \quad (0 < x \leq 1) \quad (6.8)$$

with Abel limiting at $x = 1$, and summing against $n^{-\beta}$, gives (6.4).

For the modular identity, fix $q \geq 2$. The log-sum inequality gives, for each residue class a ,

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{q}}} \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)} \geq \mu_{q,\beta}(a) \log \frac{\mu_{q,\beta}(a)}{\sum_{n \equiv a \pmod{q}} \rho_\beta(n+1)}. \quad (6.9)$$

For $a \neq 0$, the denominator is $\mu_{q,\beta}(a+1)$. For $a = 0$, it is $\mu_{q,\beta}(1) - \rho_\beta(1) \leq \mu_{q,\beta}(1)$. Summing (6.9) over a gives

$$\Sigma(\beta) \geq B_q(\beta) \quad (q \geq 2). \quad (6.10)$$

Conversely, fix $M \in \mathbb{N}$. If $q > M + 1$, then for $1 \leq a \leq M + 1$,

$$\mu_{q,\beta}(a) = \rho_\beta(a) + O_{\beta,M}(q^{-\beta}), \quad (6.11)$$

and $\mu_{q,\beta}(0) = q^{-\beta}$. All cyclic terms in $B_q(\beta)$ are nonnegative except possibly the boundary term, which satisfies

$$\mu_{q,\beta}(0) \log \frac{\mu_{q,\beta}(0)}{\mu_{q,\beta}(1)} = O_\beta(q^{-\beta} \log q). \quad (6.12)$$

Therefore

$$\liminf_{q \rightarrow \infty} B_q(\beta) \geq \sum_{n=1}^M \rho_\beta(n) \log \frac{\rho_\beta(n)}{\rho_\beta(n+1)}. \quad (6.13)$$

Letting $M \rightarrow \infty$ gives $\liminf_q B_q(\beta) \geq \Sigma(\beta)$, which together with (6.10) proves (6.5). $\square$

Corollary 6.3 (Prime-modulus Dirichlet L-resolution). *Let p be prime, let χ range over Dirichlet characters modulo p , and define*

$$S_a(p, \beta) = \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi) \quad (1 \leq a \leq p-1). \quad (6.14)$$

Then

$$\mu_{p,\beta}(a) = \frac{S_a(p, \beta)}{(p-1)\zeta(\beta)} \quad (1 \leq a \leq p-1), \quad \mu_{p,\beta}(0) = p^{-\beta}. \quad (6.15)$$

Consequently the prime-modulus functionals obtained by substituting (6.15) into $B_p(\beta)$ satisfy

$$\Sigma(\beta) = \lim_{\substack{p \rightarrow \infty \\ p \text{ prime}}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}) = \sup_{p \text{ prime}} F_p(\zeta(\beta), \{L(\beta, \chi)\}_{\chi \bmod p}). \quad (6.16)$$

Proof. For $a \not\equiv 0 \pmod{p}$, character orthogonality gives

$$\frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} \chi(n) = \begin{cases} 1, & n \equiv a \pmod{p}, \\ 0, & n \not\equiv a \pmod{p}. \end{cases} \quad (6.17)$$

Therefore

$$\sum_{\substack{n \geq 1 \\ n \equiv a \pmod{p}}} n^{-\beta} = \frac{1}{p-1} \sum_{\chi \bmod p} \overline{\chi(a)} L(\beta, \chi). \quad (6.18)$$

Dividing by $\zeta(\beta)$ gives the nonzero residue formula, while

$$\mu_{p,\beta}(0) = \frac{1}{\zeta(\beta)} \sum_{m=1}^{\infty} (mp)^{-\beta} = p^{-\beta}. \quad (6.19)$$

The functional identity follows by substituting these residue formulas into $B_p(\beta)$ and applying Theorem 6.2 along the prime sequence. $\square$

7 Euler-score identities

Theorem 7.1 (Euler-score decomposition). *For every $\beta > 1$,*

$$-(\log \zeta)'(\beta) = \mathbb{E}_\beta[\log N] = \sum_{d=1}^{\infty} \Lambda(d) \mu_{d,\beta}(0) = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}. \quad (7.1)$$

Equivalently,

$$-\frac{\zeta'(\beta)}{\zeta(\beta)} = \sum_{d=1}^{\infty} \frac{\Lambda(d)}{d^\beta}. \quad (7.2)$$

Proof. The von Mangoldt identity is

$$\log n = \sum_{d|n} \Lambda(d). \quad (7.3)$$

By nonnegative summation,

$$\mathbb{E}_\beta[\log N] = \sum_{n \geq 1} \rho_\beta(n) \sum_{d|n} \Lambda(d) = \sum_{d \geq 1} \Lambda(d) \mathbb{P}_\beta(d | N). \quad (7.4)$$

The divisibility probability is

$$\mathbb{P}_\beta(d | N) = \frac{1}{\zeta(\beta)} \sum_{m \geq 1} (dm)^{-\beta} = d^{-\beta}. \quad (7.5)$$

Substitution into (7.4) proves (7.1); (7.2) follows from (4.5). $\square$

Proposition 7.2 (Finite Euler-score identity). *For every $A \subseteq \{1, \dots, N\}$,*

$$\sum_{n \in A} \log n = \sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md). \quad (7.6)$$

Proof. Reverse the order of summation:

$$\sum_{d \leq N} \Lambda(d) \sum_{m \leq N/d} \mathbf{1}_A(md) = \sum_{n \in A} \sum_{d|n} \Lambda(d) = \sum_{n \in A} \log n. \quad (7.7)$$

$\square$

8 Curvature and Mellin-Planck tools

Lemma 8.1 (Uniform positive-axis curvature bound). *For every $s \geq 3$,*

$$(\log \zeta)''(s) = \text{Var}_s(\log N) < \frac{1}{3}. \quad (8.1)$$

Proof. By Proposition 4.3,

$$(\log \zeta)''(s) = \frac{\zeta''(s)}{\zeta(s)} - \left(\frac{\zeta'(s)}{\zeta(s)} \right)^2 \leq \frac{\zeta''(s)}{\zeta(s)} \leq \zeta''(s). \quad (8.2)$$

For $s \geq 3$,

$$\zeta''(s) = \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^s} \leq \sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3}. \quad (8.3)$$

The function $x \mapsto (\log x)^2 x^{-3}$ is decreasing for $x \geq 2$, so

$$\sum_{m=2}^{\infty} \frac{(\log m)^2}{m^3} \leq \frac{(\log 2)^2}{8} + \int_2^{\infty} \frac{(\log x)^2}{x^3} dx. \quad (8.4)$$

Since

$$\int_2^{\infty} \frac{(\log x)^2}{x^3} dx = \frac{(\log 2)^2 + \log 2 + \frac{1}{2}}{8}, \quad (8.5)$$

we obtain

$$(\log \zeta)''(s) \leq \frac{2(\log 2)^2 + \log 2 + \frac{1}{2}}{8} < \frac{1}{3}. \quad (8.6)$$

$\square$

Definition 8.2 (Mellin-Planck partition function). For $s > 1$, define

$$M(s) = \Gamma(s)\zeta(s) = \int_0^\infty \frac{t^{s-1}}{e^t - 1} dt. \quad (8.7)$$

The corresponding probability law is

$$\nu_s(dt) = \frac{t^{s-1}}{(e^t - 1)M(s)} dt. \quad (8.8)$$

Proposition 8.3 (Mellin-Planck log-convexity). For $s > 1$,

$$\frac{d^2}{ds^2} \log M(s) = \text{Var}_{\nu_s}(\log t) \geq 0. \quad (8.9)$$

Proof. Differentiate the logarithm of the integral in (8.7) under the integral sign. The first derivative is the ν_s -expectation of $\log t$, and the second derivative is its variance:

$$\frac{d^2}{ds^2} \log M(s) = \mathbb{E}_{\nu_s}[(\log t)^2] - \mathbb{E}_{\nu_s}[\log t]^2. \quad (8.10)$$

□

Bulboacă and Zayed proved several monotonicity criteria for functions of the form $u(s)/(\Gamma(s + \rho)\Gamma(s))$ and asked for best possible positive values of ρ in their examples [5]. The next application gives the sharp endpoint in the base reciprocal Gamma-product case and the corresponding endpoint criterion for monotone logarithmic derivatives.

Theorem 8.4 (APP-0009: Sharp reciprocal Gamma-product monotonicity threshold). Let $\psi = \Gamma'/\Gamma$, let γ be Euler's constant, and let $\rho_* > 0$ be the unique solution of

$$\psi(1 + \rho_*) = \gamma. \quad (8.11)$$

For $\rho > -1$, define

$$\varphi_\rho(s) = \frac{1}{\Gamma(s + \rho)\Gamma(s)}, \quad s \geq 1. \quad (8.12)$$

Then φ_ρ is strictly decreasing on $[1, \infty)$ if and only if

$$\rho \geq \rho_*. \quad (8.13)$$

Equivalently,

$$W_\rho(s) = \Gamma(s + \rho)\Gamma(s) \quad (8.14)$$

is strictly increasing on $[1, \infty)$ if and only if (8.13) holds. Numerically,

$$\rho_* = 1.2583969670859318174106234224981693941 \dots \quad (8.15)$$

More generally, let $u : [1, \infty) \rightarrow (0, \infty)$ be differentiable and suppose

$$J(s) = \frac{u'(s)}{u(s)} \quad (8.16)$$

is nonincreasing on $[1, \infty)$. Then

$$\Phi_{\rho, u}(s) = \frac{u(s)}{\Gamma(s + \rho)\Gamma(s)} \quad (8.17)$$

is strictly decreasing on $[1, \infty)$ if and only if

$$\psi(1 + \rho) \geq \gamma + J(1). \quad (8.18)$$

Proof. The digamma function is strictly increasing on $(0, \infty)$, since

$$\psi'(x) = \sum_{k=0}^{\infty} \frac{1}{(x+k)^2} > 0. \quad (8.19)$$

Hence (8.11) has a unique positive solution.

For $s \geq 1$, the logarithmic derivative of (8.14) is

$$\frac{d}{ds} \log W_{\rho}(s) = \psi(s + \rho) + \psi(s). \quad (8.20)$$

The right-hand side is strictly increasing in s . Its minimum on $[1, \infty)$ is therefore

$$\psi(1 + \rho) + \psi(1) = \psi(1 + \rho) - \gamma. \quad (8.21)$$

If $\rho \geq \rho_*$, then (8.21) is nonnegative, and it is positive for every $s > 1$. Thus, for $1 \leq a < b$,

$$\log W_{\rho}(b) - \log W_{\rho}(a) = \int_a^b (\psi(s + \rho) + \psi(s)) ds > 0. \quad (8.22)$$

So W_{ρ} is strictly increasing and $\varphi_{\rho} = 1/W_{\rho}$ is strictly decreasing. Conversely, if $\rho < \rho_*$, then (8.21) is negative, so by continuity W_{ρ} decreases initially and φ_{ρ} increases initially. This proves the base equivalence.

For the general statement,

$$\frac{d}{ds} \log \Phi_{\rho,u}(s) = J(s) - \psi(s + \rho) - \psi(s). \quad (8.23)$$

The term J is nonincreasing by hypothesis, while $\psi(s + \rho) + \psi(s)$ is strictly increasing. Therefore the logarithmic derivative in (8.23) is strictly decreasing, and its maximum on $[1, \infty)$ is

$$J(1) - \psi(1 + \rho) - \psi(1) = J(1) - \psi(1 + \rho) + \gamma. \quad (8.24)$$

The function $\Phi_{\rho,u}$ is strictly decreasing exactly when this maximum is nonpositive, which is (8.18). If the inequality fails, the logarithmic derivative is positive near $s = 1$, so monotonicity fails. $\square$

Qi, Lim, and Nantomah asked whether the second derivative of $\log \Gamma(x) + \log \Gamma(1/x)$ is completely monotonic on $(0, \infty)$ [4]. The next theorem answers this in the affirmative by a reciprocal-Weierstrass Laplace kernel.

Theorem 8.5 (APP-0004: Complete monotonicity of reciprocal-Gamma curvature). *Let*

$$H(x) = \log \Gamma(x) + \log \Gamma(1/x), \quad x > 0. \quad (8.25)$$

Then H'' is strictly completely monotonic on $(0, \infty)$; that is,

$$(-1)^m \frac{d^m}{dx^m} H''(x) > 0 \quad (m \geq 0, x > 0). \quad (8.26)$$

Proof. The Weierstrass product gives, for $x > 0$,

$$\log \Gamma(x) = -\log x - \gamma x - \sum_{n=1}^{\infty} \left(\log \left(1 + \frac{x}{n} \right) - \frac{x}{n} \right). \quad (8.27)$$

After substituting $1/x$ in (8.27), adding the two formulas, and differentiating twice on compact subintervals of $(0, \infty)$, the logarithmic second-derivative terms cancel and

$$H''(x) = \sum_{n=1}^{\infty} \frac{1}{(x+n)^2} - \frac{2\gamma}{x^3} + \sum_{n=1}^{\infty} \left[\frac{1}{(x+1/n)^2} - \frac{1}{x^2} + \frac{2}{nx^3} \right]. \quad (8.28)$$

The elementary Laplace identities

$$\frac{1}{(x+a)^2} = \int_0^{\infty} t e^{-(x+a)t} dt, \quad \frac{1}{x^2} = \int_0^{\infty} t e^{-xt} dt, \quad \frac{2a}{x^3} = a \int_0^{\infty} t^2 e^{-xt} dt \quad (8.29)$$

show that, with $a = 1/n$,

$$\frac{1}{(x+1/n)^2} - \frac{1}{x^2} + \frac{2}{nx^3} = \int_0^{\infty} e^{-xt} t \left(e^{-t/n} - 1 + \frac{t}{n} \right) dt. \quad (8.30)$$

Also,

$$\sum_{n=1}^{\infty} \frac{1}{(x+n)^2} = \int_0^{\infty} e^{-xt} \frac{t}{e^t - 1} dt, \quad (8.31)$$

and

$$-\frac{2\gamma}{x^3} = \int_0^{\infty} e^{-xt} (-\gamma t^2) dt. \quad (8.32)$$

Combining (8.28)–(8.32) gives

$$H''(x) = \int_0^{\infty} e^{-xt} K(t) dt, \quad (8.33)$$

where

$$K(t) = \frac{t}{e^t - 1} + t \sum_{n=1}^{\infty} \left(e^{-t/n} - 1 + \frac{t}{n} \right) - \gamma t^2. \quad (8.34)$$

It remains to prove $K(t) > 0$. Put

$$\phi(u) = e^{-u} - 1 + u. \quad (8.35)$$

Then $\phi \geq 0$, $\phi' = 1 - e^{-u} \geq 0$, and $u \mapsto \phi(t/u)$ is positive and decreasing on $[1, \infty)$. Therefore

$$\sum_{n=1}^{\infty} \phi(t/n) \geq \int_1^{\infty} \phi(t/u) du. \quad (8.36)$$

With $v = t/u$, integration by parts gives

$$\int_1^{\infty} \phi(t/u) du = t \int_0^t \frac{\phi(v)}{v^2} dv = -\phi(t) + t \int_0^t \frac{1 - e^{-v}}{v} dv. \quad (8.37)$$

Using

$$\int_0^t \frac{1 - e^{-v}}{v} dv = \gamma + \log t + E_1(t), \quad E_1(t) = \int_t^{\infty} \frac{e^{-v}}{v} dv, \quad (8.38)$$

we obtain

$$\frac{K(t)}{t} \geq \frac{1}{e^t - 1} - \phi(t) + t \log t + t E_1(t). \quad (8.39)$$

Since $E_1(t) > 0$ and $\log t \geq 1 - 1/t$ for $t > 0$,

$$\frac{K(t)}{t} > \frac{1}{e^t - 1} - \phi(t) + t - 1 = \frac{1}{e^t - 1} - e^{-t} = \frac{e^{-t}}{e^t - 1} > 0. \quad (8.40)$$

Thus $K(t) > 0$ for $t > 0$. Differentiating under the Laplace integral in (8.33) gives

$$(-1)^m \frac{d^m}{dx^m} H''(x) = \int_0^\infty t^m e^{-xt} K(t) dt > 0. \quad (8.41)$$

This proves (8.26). $\square$

9 External zeta inequality proofs

Theorem 9.1 (APP-0002: Affirmative solution of Nantomah zeta positivity problem). *For every $n \in \mathbb{N}$,*

$$(n+2)\zeta(n+1)\zeta(n+3) - (n+1)\zeta(n+2)^2 - \zeta(n+1)\zeta(n+2) > 0. \quad (9.1)$$

Proof. Put $s = n+1 \geq 2$ and $Z_s = \zeta(s)$. The expression in (9.1) becomes

$$K_s = (s+1)Z_s Z_{s+2} - sZ_{s+1}^2 - Z_s Z_{s+1}. \quad (9.2)$$

Let $X(m) = 1/m$ under ρ_s . By (4.51),

$$\mathbb{E}_s[X] = \frac{Z_{s+1}}{Z_s}, \quad \mathbb{E}_s[X^2] = \frac{Z_{s+2}}{Z_s}. \quad (9.3)$$

Thus

$$\frac{K_s}{Z_s^2} = (s+1)\mathbb{E}_s[X^2] - s\mathbb{E}_s[X]^2 - \mathbb{E}_s[X] = (s+1)\text{Var}_s(X) + \mathbb{E}_s[X]^2 - \mathbb{E}_s[X]. \quad (9.4)$$

Write $a = \zeta(s) - 1$, $b = \zeta(s+1) - 1$, and $c = \zeta(s+2) - 1$. Expanding gives

$$K_s = L_s + Q_s, \quad L_s = sa + (s+1)c - (2s+1)b, \quad Q_s = (s+1)ac - sb^2 - ab. \quad (9.5)$$

The linear part is termwise positive:

$$L_s = \sum_{m=2}^\infty m^{-s} \frac{(m-1)(s(m-1)-1)}{m^2} \geq 2^{-s} \frac{s-1}{4}. \quad (9.6)$$

Since $b \leq a/2$ and $c \geq 0$,

$$Q_s \geq -sb^2 - ab \geq -\frac{s+2}{4}a^2. \quad (9.7)$$

For $s \geq 4$,

$$a = \sum_{m=2}^\infty m^{-s} \leq 2^{-s} + \int_2^\infty x^{-s} dx = 2^{-s} \left(1 + \frac{2}{s-1}\right). \quad (9.8)$$

Equations (9.6), (9.7), and (9.8) imply

$$K_s \geq \frac{1}{4} \left[2^{-s}(s-1) - (s+2)2^{-2s} \left(1 + \frac{2}{s-1}\right)^2 \right]. \quad (9.9)$$

It is therefore enough to show

$$2^s(s-1) > (s+2) \left(1 + \frac{2}{s-1}\right)^2. \quad (9.10)$$

Equivalently,

$$F(s) = \frac{2^s(s-1)^3}{(s+2)(s+1)^2} > 1. \quad (9.11)$$

Now $F(4) = 72/25 > 1$, and

$$\frac{d}{ds} \log F(s) = \log 2 + \frac{3}{s-1} - \frac{1}{s+2} - \frac{2}{s+1} > 0 \quad (s \geq 4). \quad (9.12)$$

Thus $K_s > 0$ for $s \geq 4$.

For $s = 2$,

$$K_2 = 3\zeta(2)\zeta(4) - 2\zeta(3)^2 - \zeta(2)\zeta(3). \quad (9.13)$$

Using $\zeta(3) < 5/4$, $\zeta(2) = \pi^2/6$, and $\zeta(4) = \pi^4/90$,

$$K_2 > \frac{\pi^6}{180} - \frac{5\pi^2}{24} - \frac{25}{8} > \frac{10415360504209}{18000000000000} > 0. \quad (9.14)$$

For $s = 3$, the estimates $\zeta(3) > 251/216$, $\zeta(5) > 8051/7776$, and $\zeta(4) < 13/12$ yield

$$K_3 > \frac{251}{216} \left(4 \cdot \frac{8051}{7776} - \frac{13}{12}\right) - 3 \left(\frac{13}{12}\right)^2 = \frac{13783}{419904} > 0. \quad (9.15)$$

Hence $K_s > 0$ for all integers $s \geq 2$, proving the theorem. This answers Nantomah's question [1]. $\square$

Theorem 9.2 (APP-0001: Alzer-Kwong convexity and concavity pattern for reciprocal zeta). *Let $F(x) = 1/\zeta(x)$. For every integer $n \geq 1$,*

$$F''(x) > 0 \quad \text{on } (-4n, -4n+2), \quad (9.16)$$

and

$$F''(x) < 0 \quad \text{on } (-4n-2, -4n). \quad (9.17)$$

Proof. Put $x = -u$, where $u > 2$. The functional equation gives

$$\zeta(-u) = -2^{-u}\pi^{-u-1} \sin\left(\frac{\pi u}{2}\right) \Gamma(u+1)\zeta(u+1), \quad (9.18)$$

and hence

$$\frac{1}{\zeta(-u)} = -\frac{2^u\pi^{u+1}}{\Gamma(u+1)\zeta(u+1)\sin(\pi u/2)}. \quad (9.19)$$

Define the positive function

$$G(u) = \frac{2^u\pi^{u+1}}{\Gamma(u+1)\zeta(u+1)|\sin(\pi u/2)|}. \quad (9.20)$$

On $u \in (4n-2, 4n)$, $\sin(\pi u/2) < 0$, so $F(-u) = G(u)$. On $u \in (4n, 4n+2)$, $\sin(\pi u/2) > 0$, so $F(-u) = -G(u)$. Since second derivatives are unchanged by $x = -u$, it remains to prove $G''(u) > 0$.

Let $\ell(u) = \log G(u)$. Then

$$\ell(u) = u \log 2 + (u+1) \log \pi - \log \Gamma(u+1) - \log \zeta(u+1) - \log \left| \sin \left(\frac{\pi u}{2} \right) \right|. \quad (9.21)$$

Differentiating twice gives

$$\ell''(u) = -\psi'(u+1) - (\log \zeta)''(u+1) + \frac{\pi^2}{4} \csc^2 \left(\frac{\pi u}{2} \right). \quad (9.22)$$

With $s = u+1 > 3$,

$$\psi'(s) = \sum_{k=0}^{\infty} \frac{1}{(s+k)^2} < \int_{s-1}^{\infty} \frac{dt}{t^2} = \frac{1}{s-1} \leq \frac{1}{2}. \quad (9.23)$$

Combining (8.1), (9.23), and $\csc^2(\pi u/2) \geq 1$, we get

$$\ell''(u) > \frac{\pi^2}{4} - \frac{1}{2} - \frac{1}{3} = \frac{\pi^2}{4} - \frac{5}{6} > 0. \quad (9.24)$$

Since $G > 0$,

$$G''(u) = G(u)(\ell''(u) + \ell'(u)^2) > 0. \quad (9.25)$$

The two signs in (9.16) and (9.17) now follow from $F(-u) = G(u)$ on $u \in (4n-2, 4n)$ and $F(-u) = -G(u)$ on $u \in (4n, 4n+2)$. This proves the Alzer–Kwong sign pattern [3]. $\square$

Theorem 9.3 (APP-0003: Generalized Holder inequality for Gamma zeta). *Let $m \geq 2$, $r \geq 1$, $x_1, \dots, x_m > 1$, and $p_1, \dots, p_m > 1$ satisfy*

$$\sum_{i=1}^m \frac{1}{p_i} = \frac{1}{r}. \quad (9.26)$$

Define

$$T = r \sum_{i=1}^m \frac{x_i}{p_i}. \quad (9.27)$$

Then $T > 1$ and

$$\zeta(T) \leq \frac{\prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}}{\Gamma(T)}. \quad (9.28)$$

Proof. Set

$$\lambda_i = \frac{r}{p_i}. \quad (9.29)$$

Then $\lambda_i > 0$, $\sum_i \lambda_i = 1$, and $T = \sum_i \lambda_i x_i > 1$. Define

$$q_i = \frac{p_i}{r}. \quad (9.30)$$

Then $\sum_i 1/q_i = 1$, and the assumptions imply $q_i > 1$ for all i ; otherwise one term $1/p_i$ would already be at least $1/r$, leaving no room for the remaining positive terms in (9.26).

For $t > 0$, let

$$f_i(t) = \left(\frac{t^{x_i-1}}{e^t - 1} \right)^{1/p_i}. \quad (9.31)$$

Then

$$\left(\prod_{i=1}^m f_i(t) \right)^r = \frac{t^{r \sum_i (x_i - 1)/p_i}}{e^t - 1} = \frac{t^{T-1}}{e^t - 1}. \quad (9.32)$$

Consequently,

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r(0, \infty)}^r = \int_0^\infty \frac{t^{T-1}}{e^t - 1} dt = \Gamma(T) \zeta(T). \quad (9.33)$$

Generalized Holder gives

$$\left\| \prod_{i=1}^m f_i \right\|_{L^r} \leq \prod_{i=1}^m \|f_i\|_{L^{p_i}}. \quad (9.34)$$

Moreover,

$$\|f_i\|_{L^{p_i}}^{p_i} = \int_0^\infty \frac{t^{x_i-1}}{e^t - 1} dt = \Gamma(x_i) \zeta(x_i). \quad (9.35)$$

Combining (9.33), (9.34), and (9.35), then raising to the power r , gives

$$\Gamma(T) \zeta(T) \leq \prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}. \quad (9.36)$$

Division by $\Gamma(T)$ proves (9.28). This solves the generalized Holder problem in Sroysang's notation, where his $\xi(s)$ is $\zeta(s)$ for $s > 1$ [2]. $\square$

Remark 9.4 (Free-energy interpretation of the Sroysang inequality). The same result follows from log-convexity of $M(s) = \Gamma(s) \zeta(s)$. With $\lambda_i = r/p_i$ and $T = \sum_i \lambda_i x_i$, Jensen's inequality for $\log M$ gives

$$M(T) \leq \prod_{i=1}^m M(x_i)^{\lambda_i} = \prod_{i=1}^m (\Gamma(x_i) \zeta(x_i))^{r/p_i}, \quad (9.37)$$

which is equivalent to (9.28).

10 Laplace moment-ratio application

The complete-monotonicity proofs above repeatedly convert special-function curvature questions into positivity statements for Laplace kernels. The next lemma records a reusable moment-ratio consequence of that viewpoint.

Lemma 10.1 (Laplace moment-ratio bridge). *Let μ be a positive Borel measure on $[0, \infty)$, and assume that the tilted moments*

$$a_j(x) = \int_0^\infty t^j e^{-xt} d\mu(t) \quad (10.1)$$

are finite and positive for the displayed indices. Then, for every $n \geq 0$,

$$B_n(x) = \frac{a_{n+1}(x)}{a_n(x) a_{n+2}(x)} \quad (10.2)$$

is strictly increasing in x .

Proof. Put $r_j(x) = a_{j+1}(x)/a_j(x)$. Cauchy–Schwarz gives

$$a_{j+1}(x)^2 \leq a_j(x)a_{j+2}(x), \quad (10.3)$$

so $r_j(x) \leq r_{j+1}(x)$. Since $a'_j(x) = -a_{j+1}(x)$, differentiating (10.2) logarithmically gives

$$\frac{d}{dx} \log B_n(x) = -\frac{a_{n+2}}{a_{n+1}} + \frac{a_{n+1}}{a_n} + \frac{a_{n+3}}{a_{n+2}} = r_n(x) - r_{n+1}(x) + r_{n+2}(x). \quad (10.4)$$

The right-hand side is at least $r_n(x) > 0$, because $r_{n+2}(x) \geq r_{n+1}(x)$. Hence B_n is strictly increasing. $\square$

Yin and Zhang define the Nielsen k -beta function by

$$\beta_k(x) = \int_0^1 \frac{t^{x-1}}{1+t^k} dt = \int_0^\infty \frac{e^{-xt}}{1+e^{-kt}} dt, \quad k > 0, x > 0. \quad (10.5)$$

They ask for the monotonicity of the derivative ratio involving $x\beta_k(x)$ [8]. The bridge lemma gives the exact parity law.

Theorem 10.2 (APP-0010: Nielsen k -beta derivative-ratio monotonicity). *Let $k > 0$, let $n \geq 0$, and put $f_k(x) = x\beta_k(x)$. Then*

$$R_{k,n}(x) = \frac{f_k^{(n+1)}(x)}{f_k^{(n)}(x)f_k^{(n+2)}(x)} \quad (10.6)$$

is strictly increasing on $(0, \infty)$ when n is odd, and strictly decreasing on $(0, \infty)$ when n is even.

Proof. From (10.5),

$$f_k(x) = x \int_0^\infty \frac{e^{-xt}}{1+e^{-kt}} dt. \quad (10.7)$$

Integrating by parts gives

$$f_k(x) = \frac{1}{2} + \int_0^\infty e^{-xt} \frac{ke^{-kt}}{(1+e^{-kt})^2} dt. \quad (10.8)$$

Thus f_k is a Laplace transform of a positive measure consisting of an atom $1/2$ at 0 and a positive density on $(0, \infty)$. Define

$$a_j(x) = (-1)^j f_k^{(j)}(x). \quad (10.9)$$

For the needed indices these are positive tilted moments of the measure in (10.8). Lemma 10.1 shows that

$$\frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)} \quad (10.10)$$

is strictly increasing. But

$$R_{k,n}(x) = (-1)^{n+1} \frac{a_{n+1}(x)}{a_n(x)a_{n+2}(x)}. \quad (10.11)$$

Multiplication by $(-1)^{n+1}$ gives strict increase for odd n and strict decrease for even n . This resolves the Nielsen k -beta derivative-ratio problem with the precise parity convention. $\square$

11 Endpoint and parameter applications

The next three applications add one certified endpoint result and two parameter-monotonicity results to the complete-monotonicity layer. Only fully solved source questions are labelled.

11.1 Exact $n = 2$ beta-window endpoint

For $s > 1$, write

$$Z_s(a) = \sum_{m=0}^{\infty} (m+a)^{-s}. \quad (11.1)$$

In Qi–Lim–Nantomah’s notation [4], put

$$C_2(x) = \psi''(x) + x\psi^{(3)}(x), \quad P_2(x) = \psi''(x)\psi''(1/x), \quad (11.2)$$

and define

$$\mathcal{I}_2 = \{\beta \in \mathbb{R} : x^\beta C_2(x) - P_2(x) < 0 \text{ for all } x > 0\}. \quad (11.3)$$

The remaining endpoint is controlled on $0 < x < 1$ by

$$R(x) = \frac{P_2(x)}{C_2(x)} = \frac{2Z_3(x)Z_3(1/x)}{3xZ_4(x) - Z_3(x)}, \quad Q_2(x) = \frac{\log R(x)}{\log x}. \quad (11.4)$$

Theorem 11.1 (APP-0011: exact $n = 2$ beta-window endpoint). *Let*

$$J = \left[\frac{287345}{1000000}, \frac{287346}{1000000} \right]. \quad (11.5)$$

There is a unique $\xi \in J$ at which Q_2 attains its global supremum on $(0, 1)$. Hence, with $L_2 = Q_2(\xi)$,

$$\mathcal{I}_2 = (L_2, 3]. \quad (11.6)$$

Proof. The identity $\psi^{(m)}(x) = (-1)^{m+1}m!Z_{m+1}(x)$ gives (11.4). Define

$$\Lambda(x) = -\frac{3Z_4(x)}{Z_3(x)} + \frac{3Z_4(1/x)}{x^2Z_3(1/x)} - \frac{6Z_4(x) - 12xZ_5(x)}{3xZ_4(x) - Z_3(x)} \quad (11.7)$$

and

$$G(x) = x \log x \Lambda(x) - \log R(x). \quad (11.8)$$

Differentiating (11.4) gives

$$Q_2'(x) = \frac{G(x)}{x(\log x)^2}. \quad (11.9)$$

All inequalities below are rational interval certificates using the Hurwitz-zeta integral-tail enclosure and the atanh logarithm enclosure. First,

$$G\left(\frac{287345}{1000000}\right) > 0, \quad G\left(\frac{287346}{1000000}\right) < 0. \quad (11.10)$$

Second, on

$$B = \left[\frac{1409}{5000}, \frac{293}{1000} \right],$$

the same enclosure proves

$$H(x) = \Lambda(x) + x\Lambda'(x) > 0. \quad (11.11)$$

Since $G'(x) = \log x H(x)$ and $\log x < 0$ on B , G is strictly decreasing on B . Together with (11.10), this gives a unique zero $\xi \in J$, and (11.9) shows that Q_2 is increasing before ξ and decreasing after ξ on B .

It remains only to exclude points outside the compact bracket. The endpoint certificate gives a rational lower witness $q_J < Q_2(\xi)$ with

$$\frac{115727}{50000} < q_J. \quad (11.12)$$

On $(0, 9/50]$ the scaled near-zero bound gives $Q_2(x) < 23/10 < q_J$. On $[9/10, 1)$, a finite rational subdivision proves $R(x) > 1$, hence $Q_2(x) < 0 < q_J$. On

$$\left[\frac{9}{50}, \frac{1409}{5000} \right] \cup \left[\frac{293}{1000}, \frac{9}{10} \right],$$

a finite rational cover proves $Q_2(x) < q_J$: the left bridge uses

$$R(x) > Z_3(1/x) > x^{1157/500},$$

and the remaining intervals use the direct comparison

$$\log R(x) > \frac{115727}{50000} \log x.$$

Thus no point outside B reaches the value at ξ , so $L_2 = Q_2(\xi)$.

The source reduction gives the upper endpoint 3 and the convention that the lower endpoint is excluded when equality occurs. At $\beta = L_2$, equality occurs at $x = \xi$; therefore the exact admissible set is $(L_2, 3]$. $\square$

11.2 Baricz V_q parameter log-convexity

Baricz records the question whether, for fixed $x > 0$, the function

$$V_q(x) = \frac{2e^{x^2}}{\Gamma(q+1)} \int_x^\infty e^{-t^2} (t^2 - x^2)^q dt, \quad q > -1, \quad (11.13)$$

is log-convex as a function of q [9].

Theorem 11.2 (APP-0012: Baricz V_q strict q -log-convexity). *For each fixed $x > 0$, the map $q \mapsto V_q(x)$ is strictly log-convex on $(-1, \infty)$.*

Proof. Put $u = t^2 - x^2$. Then

$$V_q(x) = \frac{1}{\Gamma(q+1)} \int_0^\infty e^{-u} u^q (u + x^2)^{-1/2} du. \quad (11.14)$$

Using

$$r^{-1/2} = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-rs} ds$$

and Tonelli's theorem, we obtain

$$V_q(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty s^{-1/2} e^{-x^2 s} (1+s)^{-(q+1)} ds. \quad (11.15)$$

Let $a = q + 1 > 0$. Differentiation under the integral gives

$$(-1)^n \frac{d^n}{da^n} V_{a-1}(x) = \frac{1}{\sqrt{\pi}} \int_0^\infty (\log(1+s))^n s^{-1/2} e^{-x^2 s} (1+s)^{-a} ds \geq 0. \quad (11.16)$$

Equivalently, after the change of variables $y = \log(1+s)$, $V_{a-1}(x)$ is a positive Laplace transform in a . The representing measure is not concentrated at a single point, because the density in (11.15) is positive on $s > 0$. Strict Holder inequality for two different exponents therefore gives strict log-convexity in a , and the shift $a = q + 1$ gives the stated strict log-convexity in q . $\square$

11.3 Yin (p, k) -digamma sharp necessity

Yin proved sufficiency of $\alpha \leq 1$ for complete monotonicity of

$$\delta_{p,k,\alpha}(x) = x^\alpha \left[\frac{1}{k} \log \frac{pkx}{x + k(p+1)} - \psi_{p,k}(x) \right], \quad p \in \mathbb{N}, \quad k > 0, \quad (11.17)$$

and asked whether this condition is also necessary [10].

Theorem 11.3 (APP-0013: Yin (p, k) -digamma sharp α -necessity). *If $\delta_{p,k,\alpha}$ is completely monotone on $(0, \infty)$, then $\alpha \leq 1$. Combined with Yin's sufficiency theorem, the sharp condition is $\alpha \leq 1$.*

Proof. The source representation is

$$\psi_{p,k}(x) = \frac{1}{k} \log(pk) - \sum_{n=0}^p \frac{1}{x + nk}. \quad (11.18)$$

Therefore the bracket in (11.17) is

$$B_{p,k}(x) = \frac{1}{k} \log \frac{x}{x + k(p+1)} + \sum_{n=0}^p \frac{1}{x + nk}. \quad (11.19)$$

With $t = x/k$,

$$B_{p,k}(x) = \frac{1}{k} \left[\sum_{n=0}^p \frac{1}{t + n} + \log \frac{t}{t + p + 1} \right].$$

Since $u \mapsto (t + u)^{-1}$ is strictly decreasing,

$$\sum_{n=0}^p \frac{1}{t + n} > \int_0^{p+1} \frac{du}{t + u} = \log \frac{t + p + 1}{t},$$

so $B_{p,k}(x) > 0$ for $x > 0$.

As $x \rightarrow 0^+$, (11.19) gives

$$B_{p,k}(x) = \frac{1}{x} + O(|\log x|), \quad \delta_{p,k,\alpha}(x) \sim x^{\alpha-1}. \quad (11.20)$$

If $\alpha > 1$, then $\delta_{p,k,\alpha}(x) \rightarrow 0$ as $x \rightarrow 0^+$. But a completely monotone nonzero function is positive and nonincreasing on $(0, \infty)$. For any fixed $x_0 > 0$, positivity gives $\delta_{p,k,\alpha}(x_0) > 0$, while the limit at 0 gives a point $0 < y < x_0$ with $\delta_{p,k,\alpha}(y) < \delta_{p,k,\alpha}(x_0)$, contradicting nonincreasingness. Hence $\alpha \leq 1$. $\square$

12 The Laplace-Transport Application Layer

The applications in this layer are organized by one theme. A problem is first converted into a positive-kernel statement, usually by a logarithmic second difference, a Frullani identity, a moment determinant, or a transform normalization. The conclusion then follows by transporting that kernel through a permanence operation: Stieltjes transformation, Bernstein integration, stable subordination, quotient compression, or triangular coefficient extraction. This section records representative public solves.

Theorem 12.1 (APP-0014: Ramanujan integral is Stieltjes). *Let*

$$I_R(x) = \int_0^\infty \frac{e^{-xt}}{t(\pi^2 + \log^2 t)} dt, \quad x > 0.$$

Then I_R is a Stieltjes function. Moreover every primitive of the form

$$\tilde{I}_R(x) = a + \int_0^\infty \frac{1 - e^{-xt}}{t(\pi^2 + \log^2 t)} dt, \quad a \geq 0,$$

is a complete Bernstein function.

Proof. For $t > 0$,

$$\frac{1}{t(\pi^2 + \log^2 t)} = \frac{1}{\pi(1+t)} \int_0^1 t^{-u} \sin(\pi u) du.$$

The right side is a positive superposition of completely monotone powers and is therefore completely monotone as a function of t . The Laplace transform of a completely monotone density is Stieltjes. Integrating the same density against $1 - e^{-xt}$ gives a complete Bernstein function by the standard Stieltjes-density characterization of complete Bernstein functions. $\square$

Theorem 12.2 (APP-0015: Prabhakar Q-measure by stable subordination). *Let $0 < \alpha < 1$, $\eta > 0$, $\beta > \alpha\eta$, and $\lambda > 0$. Let*

$$E_{\alpha,\beta}^\eta(z) = \sum_{m=0}^\infty \frac{(\eta)_m z^m}{m! \Gamma(\alpha m + \beta)}$$

be the Prabhakar Mittag-Leffler function. Let $P_{\alpha,\beta}^\eta$ be the transform-normalized Pollard measure characterized in the strict Pollard range by

$$\int_0^\infty e^{-xr} dP_{\alpha,\beta}^\eta(r) = E_{\alpha,\beta}^\eta(-x), \quad x > 0.$$

Let S_α be the positive α -stable variable satisfying $\mathbb{E}e^{-sS_\alpha} = e^{-s^\alpha}$. Define the finite measure

$$Q_{\alpha,\beta}^\eta(A \mid \lambda) = \int \Pr\{(\lambda r)^{1/\alpha} S_\alpha \in A\} dP_{\alpha,\beta}^\eta(r).$$

Then

$$\int_0^\infty e^{-xt} dQ_{\alpha,\beta}^\eta(t \mid \lambda) = E_{\alpha,\beta}^\eta(-\lambda x^\alpha).$$

Proof. Conditioning on r and using the stable transform gives

$$\mathbb{E}e^{-x(\lambda r)^{1/\alpha} S_\alpha} = e^{-\lambda r x^\alpha}.$$

Integration against $P_{\alpha,\beta}^\eta$ gives the Pollard representation of $E_{\alpha,\beta}^\eta(-\lambda x^\alpha)$. The total mass is finite, equal to the value at $x = 0$, and no normalization to a probability measure is needed for the transform identity. $\square$

Theorem 12.3 (APP-0016: Mills-ratio bound at every derivative level). *Let*

$$M_L(t) = \int_0^\infty u^L e^{-tu - u^2/2} du, \quad L = 0, 1, 2, \dots$$

Put $m_L(t) = M_{L+1}(t)/M_L(t)$ and

$$U_L(t) = \frac{\sqrt{t^2(t^2 + 4L + 7)^2 + 4(L + 1)(t^2 + 4L + 8)(t^2 + 4L + 6)} - t(t^2 + 4L + 7)}{2(t^2 + 4L + 8)}.$$

Then $m_L(t) < U_L(t)$ for every $L \geq 0$ and $t > 0$. With

$$P_0 = 1, \quad P_1 = t, \quad P_{n+1} = tP_n + nP_{n-1},$$

and

$$B_0 = 0, \quad B_1 = 1, \quad B_{n+1} = nB_{n-1} - tB_n,$$

the classical Mills ratio $r(t) = M_0(t)$ satisfies

$$r(t) > \frac{B_{L+1} - U_L B_L}{P_{L+1} + U_L P_L} \quad (L \geq 0 \text{ even}),$$

and

$$r(t) < \frac{U_L B_L - B_{L+1}}{P_{L+1} + U_L P_L} \quad (L \geq 1 \text{ odd}).$$

Proof. The density defining M_L is a positive moment density. Its order-four Hankel-minor consequence gives

$$M_{L+4}M_L - 4M_{L+3}M_{L+1} + 3M_{L+2}^2 > 0.$$

Using the recurrence $M_{n+1} = nM_{n-1} - tM_n$, this determinant inequality reduces to a quadratic inequality for m_L ; solving the admissible positive root gives $m_L < U_L$. Iterating the same recurrence expresses M_L in the form

$$M_L = (-1)^L (P_L r - B_L),$$

and substituting the level- L quotient bound gives the displayed alternating upper and lower bounds for r . $\square$

Theorem 12.4 (APP-0017: a gamma quotient need not be Bernstein). *The assertion that the gamma quotient*

$$x \mapsto \frac{\Gamma(x+a)\Gamma(x+b)}{\Gamma(x)\Gamma(x+a+b)}$$

is Bernstein for all positive a, b is false.

Proof. Take $a = 2$, $b = 3$, and set $y = x - 2$. The quotient becomes

$$g(y) = \frac{y(y+1)}{(y+3)(y+4)}.$$

A direct differentiation gives

$$g'''(y) = \frac{36(y^4 + 8y^3 + 12y^2 - 40y - 94)}{(y+3)^4(y+4)^4}.$$

Thus $g'''(1) < 0$. If g were Bernstein, then g' would be completely monotone and hence $g''' \geq 0$. The displayed value contradicts that necessary condition. $\square$

Theorem 12.5 (APP-0018: exact supremum for the Qi-Guo tau threshold). *For $s \in \mathbb{N}$ and $t > 0$, define*

$$\tau(s, t) = \frac{1}{s} \left[t - (t + s + 1) \left(\frac{t}{t+1} \right)^{s+1} \right].$$

Let $a_ > 0$ be the unique positive solution of*

$$e^{a_*} = 1 + a_* + a_*^2.$$

Then

$$\sup_{s \in \mathbb{N}, t > 0} \tau(s, t) = \frac{a_*}{1 + a_* + a_*^2} = 0.298425607525639 \dots$$

The supremum is not attained.

Proof. Writing $u = t/(t+1)$ gives a two-parameter expression on $0 < u < 1$. The Euler-Lagrange equations reduce the interior extremal sequence to the scaling limit $u = e^{-a/s}$. The limiting profile is maximized exactly when $e^a = 1 + a + a^2$, and the resulting value is $a/(1 + a + a^2)$. The source one-variable monotonicity estimates bound the finite- s expressions by this limit, while the scaling sequence approaches it. Hence the number is a supremum and is not attained at finite s, t . $\square$

Theorem 12.6 (APP-0019: exact Bernstein range for the W-symbol). *Let $0 < \alpha < 1$, $\beta \geq 0$, and*

$$\Phi_{\alpha, \beta}(s) = s^\alpha (1 + (1 - \alpha)s^{\alpha-1})^{-\beta}.$$

Then $\Phi_{\alpha, \beta}$ is a Bernstein function if and only if

$$0 \leq \beta \leq 1.$$

Proof. Wakrim's source proves the non-Bernstein obstruction for $\beta > 1$ and leaves the range $0 < \beta \leq 1$ as the remaining delicate case. We prove that remaining Bernstein half in detail. No complete-Bernstein or subordination conclusion is asserted here.

Put

$$a = 1 - \alpha \in (0, 1), \quad y = s^a.$$

Since $\alpha = 1 - a$,

$$\Phi_{\alpha, \beta}(s) = s^{1-a} (1 + as^{-a})^{-\beta} = s^{1-a+a\beta} (s^a + a)^{-\beta}.$$

Let

$$p = 1 - a + a\beta = \alpha + a\beta.$$

Differentiating the last expression gives

$$\begin{aligned} \Phi'_{\alpha, \beta}(s) &= s^{p-1} (s^a + a)^{-\beta-1} \{p(s^a + a) - \beta a s^a\} \\ &= s^{p-1} (s^a + a)^{-\beta-1} \{\alpha s^a + \alpha a + a^2 \beta\} \\ &= s^{p-1} (s^a + a)^{-\beta} \left(\alpha + \frac{a^2 \beta}{s^a + a} \right). \end{aligned}$$

Because $p - 1 = a(\beta - 1)$, this is

$$\Phi'_{\alpha, \beta}(s) = H_{\alpha, \beta}(s^a),$$

where

$$H_{\alpha, \beta}(y) = y^{\beta-1} (y + a)^{-\beta} \left(\alpha + \frac{a^2 \beta}{y + a} \right).$$

We now check complete monotonicity of $H_{\alpha,\beta}$ for $0 \leq \beta \leq 1$. The first factor is

$$y^{\beta-1} = y^{-(1-\beta)},$$

and this is interpreted as follows. If $0 \leq \beta < 1$, put $r = 1 - \beta > 0$. Then

$$y^{\beta-1} = y^{-r} = \frac{1}{\Gamma(r)} \int_0^\infty e^{-yt} t^{r-1} dt,$$

so $y^{\beta-1}$ is completely monotone by the Bernstein–Widder criterion. If $\beta = 1$, the same factor is the constant 1. The shifted factor $(y+a)^{-\beta}$ is completely monotone for $\beta > 0$ by the Laplace representation

$$(y+a)^{-\beta} = \frac{1}{\Gamma(\beta)} \int_0^\infty e^{-yt} e^{-at} t^{\beta-1} dt,$$

and for $\beta = 0$ it is the constant 1. The last factor

$$\alpha + \frac{a^2\beta}{y+a}$$

is a positive constant plus a nonnegative multiple of the completely monotone resolvent $(y+a)^{-1}$. By closure of completely monotone functions under products and nonnegative sums, $H_{\alpha,\beta}$ is completely monotone.

The endpoint $\beta = 1$ has no limiting ambiguity; explicitly

$$H_{\alpha,1}(y) = \frac{\alpha}{y+a} + \frac{a^2}{(y+a)^2},$$

which is completely monotone. The endpoint $\beta = 0$ also agrees with the direct identity $\Phi_{\alpha,0}(s) = s^\alpha$.

Finally, let $g(s) = s^a$. The composition theorem used here is the standard one: if f is completely monotone on $(0, \infty)$ and g is a Bernstein function with $g((0, \infty)) \subset (0, \infty)$, then $f \circ g$ is completely monotone. Its hypotheses are satisfied in the present case: $H_{\alpha,\beta}$ is completely monotone by the factor check above, $g(s) = s^a > 0$, and g is a Bernstein function for $0 < a < 1$ since

$$g'(s) = as^{a-1} = as^{-(1-a)}$$

is completely monotone. Hence $\Phi'_{\alpha,\beta} = H_{\alpha,\beta} \circ g$ is completely monotone for $0 \leq \beta \leq 1$, so $\Phi_{\alpha,\beta}$ is a Bernstein function. Together with Wakrim's source obstruction for $\beta > 1$, this closes the remaining case and gives the exact Bernstein range. $\square$

Theorem 12.7 (APP-0020: Qi h_λ degree-four conjecture is false). *Let*

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \quad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\text{cm}} h_\lambda = \deg_x^{\text{cm}}(-h_\mu) = 4$ exactly in the stated parameter ranges $\lambda \leq 0$, $\mu \geq 4$, is false. Here $\deg_x^{\text{cm}} g$ denotes the source convention: the supremal power r for which $x^r g(x)$ is completely monotone on $(0, \infty)$.

Proof. Near zero,

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right) \frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then $x^4 h_\lambda(x) = (-\lambda/12)x + O(x^2)$, whose derivative is positive near zero. If $\lambda = 0$, the coefficient $\pi^2/3 - 37/12$ is positive, again giving a positive derivative for the degree-four transform near zero. Complete monotonicity would require the first derivative to be non-positive. The same small- x test applied to $-x^4 h_\mu$ excludes the claimed μ -range. Hence the degree-four conjecture fails. $\square$

Theorem 12.8 (APP-0021: Szabo's cutoff is $\alpha_0 = 1$). *For $0 < d < 1$, define*

$$H_d(y) = \psi(y+d) - \psi(y) - \frac{d}{y}.$$

The exact cutoff for complete monotonicity of $y^\alpha H_d(y)$ is

$$\alpha_0 = 1.$$

Proof. The source theorem proves complete monotonicity for $\alpha \leq 1$. Conversely, as $y \downarrow 0$,

$$H_d(y) = \frac{1-d}{y} + O(1).$$

Hence, for $\alpha > 1$,

$$y^\alpha H_d(y) = (1-d)y^{\alpha-1} + O(y^\alpha),$$

whose derivative is positive near zero. A completely monotone function must be non-increasing, so no $\alpha > 1$ is admissible. $\square$

Theorem 12.9 (APP-0022: Chen-Choi Gurland ratio is log-completely monotone). *Let*

$$F(x) = \frac{\Gamma(1/x)\Gamma(3/x)}{\Gamma(2/x)^2}, \quad x > 0.$$

Then F is logarithmically completely monotone; equivalently,

$$(-1)^n (\log F)^{(n)}(x) > 0, \quad n = 0, 1, 2, \dots$$

Proof. Set

$$A(u) = 2 \log(u+2) - \log(u+1) - \log(u+3).$$

The identity

$$A(u) = \int_0^\infty e^{-ut} \frac{e^{-t}(1-e^{-t})^2}{t} dt$$

shows that A is strictly completely monotone. The Weierstrass product for Γ gives

$$\log F(x) = \log \frac{4}{3} + \sum_{m \geq 1} A(mx).$$

Termwise differentiation is justified on compact subintervals of $(0, \infty)$, and each differentiated summand has the required strict alternating sign. This proves logarithmic complete monotonicity. $\square$

Theorem 12.10 (APP-0023: Sokal lambda-derivatives have triangular compression). *Let $n, k \in \mathbb{N}_0$, $\lambda > 0$, $x > 0$, and let f be a C^∞ function on $(0, \infty)$. In Sokal's generalized-Stieltjes application, the quantities below are the Hankel-type expressions associated with a generalized Stieltjes function of order λ . For $j \geq 0$, set*

$$G_j(x) = (-1)^n x^j f^{(n+j)}(x), \quad a = n + \lambda.$$

For $0 \leq r \leq k$, define

$$F_{n,r}^{[\lambda]}(x) = \sum_{j=0}^r \binom{r}{j} (a+j)_{r-j} G_j(x).$$

Then, for every integer $0 \leq \ell \leq k$,

$$\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x) = \sum_{r=0}^{k-\ell} \frac{k!}{r!} [z^{k-r}] (-\log(1-z))^\ell F_{n,r}^{[\lambda]}(x).$$

For $\ell > k$, $\partial_\lambda^\ell F_{n,k}^{[\lambda]}(x) = 0$. The coefficients in this triangular compression are non-negative.

Proof. The exponential generating function is

$$\mathcal{F}(z) = (1-z)^{-a} \sum_{j \geq 0} G_j(x) \left(\frac{z}{1-z} \right)^j \frac{1}{j!}.$$

Differentiating ℓ times with respect to λ multiplies $\mathcal{F}$ by $(-\log(1-z))^\ell$. Extracting coefficients gives the displayed triangular formula. Since $-\log(1-z) = \sum_{m \geq 1} z^m/m$ has non-negative coefficients, all compression coefficients are non-negative. $\square$

Theorem 12.11 (APP-0024: Simon gamma quotient is Bernstein). *For $0 < \alpha < 1$, define*

$$F_\alpha(x) = \frac{\Gamma(x+\alpha)}{\Gamma(x)x^\alpha}, \quad x > 0.$$

Then F_α is a Bernstein function. More precisely, $1 - F_\alpha$ is completely monotone.

Proof. Put $\beta = 1 - \alpha$. The beta integral gives

$$F_\alpha(x) = x^\beta \frac{\Gamma(x+\alpha)}{\Gamma(x+1)} = \frac{x^\beta}{\Gamma(\beta)} \int_0^\infty e^{-xu} (e^u - 1)^{-\alpha} du.$$

Define

$$J_\alpha(t) = \int_0^t (t-u)^{\alpha-1} (e^u - 1)^{-\alpha} du.$$

Laplace convolution gives

$$F_\alpha(x) = \frac{x}{\Gamma(\alpha)\Gamma(1-\alpha)} \int_0^\infty e^{-xt} J_\alpha(t) dt.$$

After the substitution $u = tv$,

$$J_\alpha(t) = \int_0^1 (1-v)^{\alpha-1} \left(\frac{t}{e^{tv} - 1} \right)^\alpha dv.$$

For each $v > 0$, the function $t/(e^{tv} - 1)$ is decreasing in t . The integrand is dominated near the endpoints by $(1 - v)^{\alpha-1}v^{-\alpha}$, which is integrable because $0 < \alpha < 1$. Hence J_α is decreasing, with

$$J_\alpha(0+) = \Gamma(\alpha)\Gamma(1 - \alpha), \quad J_\alpha(\infty) = 0.$$

Let $\mu_\alpha = -dJ_\alpha$. This is a positive finite measure, and Stieltjes integration by parts yields

$$1 - F_\alpha(x) = \frac{1}{\Gamma(\alpha)\Gamma(1 - \alpha)} \int_0^\infty e^{-xt} \mu_\alpha(dt).$$

Thus $1 - F_\alpha$ is completely monotone. Differentiating this Laplace representation gives

$$F'_\alpha(x) = \frac{1}{\Gamma(\alpha)\Gamma(1 - \alpha)} \int_0^\infty te^{-xt} \mu_\alpha(dt),$$

so F'_α is completely monotone. Therefore F_α is a Bernstein function. $\square$

Theorem 12.12 (APP-0025: Du–Wang h_3 monotonicity classification). *For $0 < a < 2$, define*

$$h_3(x) = \frac{-x^2\psi'(x+a) + 2x\psi(x+a) - 2\log\Gamma(x+a)}{x}, \quad x > 0.$$

Then h_3 is increasing on $(0, \infty)$ if and only if

$$\frac{1}{2} \leq a \leq 1.$$

For $0 < a < 1/2$ and for $1 < a < 2$, h_3 is not monotone on $(0, \infty)$.

Proof. The source reduction writes $t = x + a$ and

$$h'_3(x) = \frac{h_{31}(x+a)}{x^2},$$

where

$$h_{31}(t) = -(t-a)^3\psi''(t) + (t-a)^2\psi'(t) - 2(t-a)\psi(t) + 2\log\Gamma(t).$$

For $u > 0$, set $H_a(u) = h_{31}(a+u)$. Then

$$H'_a(u) = -u^2\{u\psi'''(a+u) + 2\psi''(a+u)\}, \quad H_a(0+) = 2\log\Gamma(a).$$

We first prove the middle window. For $t > 1/2$, put

$$S_m(t) = \sum_{n=0}^{\infty} (n+t)^{-m}.$$

Since $\psi''(t) = -2S_3(t)$ and $\psi'''(t) = 6S_4(t)$, it is enough to show

$$2S_3(t) - 3\left(t - \frac{1}{2}\right)S_4(t) > 0.$$

Let $y = t - 1/2$ and $K(x) = 1/(2\sinh(x/2))$. The standard integral representation for S_m , followed by integration by parts, gives

$$\begin{aligned} 2S_3(t) - 3yS_4(t) &= \int_0^\infty e^{-yx} \left(x^2K(x) - \frac{1}{2}(x^3K(x))' \right) dx \\ &= \int_0^\infty e^{-yx} \frac{x^2K(x)}{2} \left(\frac{x}{2} \coth \frac{x}{2} - 1 \right) dx. \end{aligned}$$

The last integrand is positive because $z \coth z > 1$ for $z > 0$. Therefore

$$-\frac{2\psi''(t)}{\psi'''(t)} > t - \frac{1}{2}.$$

If $1/2 \leq a \leq 1$, then $u = t - a \leq t - 1/2$, so

$$2\psi''(t) + u\psi'''(t) < 0.$$

Thus $H'_a(u) > 0$. Also $H_a(0+) = 2 \log \Gamma(a) \geq 0$ on $[1/2, 1]$, since $\log \Gamma$ decreases there and $\Gamma(1) = 1$. Hence $H_a(u) \geq 0$, and therefore $h'_3(x) \geq 0$, on the whole middle window.

It remains to exclude the outer windows. As $t \rightarrow \infty$, the standard asymptotics for ψ, ψ', ψ'' , and $\log \Gamma$ give

$$h_{31}(t) = (2a - 1) \log t + O(1).$$

If $0 < a < 1/2$, then $h_{31}(a+) = 2 \log \Gamma(a) > 0$, while $h_{31}(t) < 0$ for all sufficiently large t . If $1 < a < 2$, then $h_{31}(a+) = 2 \log \Gamma(a) < 0$, while $h_{31}(t) > 0$ for all sufficiently large t . In both cases h'_3 changes sign, so h_3 is not monotone. $\square$

Theorem 12.13 (APP-0026: Baskakov even-power conjecture is false). *For the Abel–Gawronski–Neuschel higher-power Baskakov family*

$$f_\alpha^{[r]}(x) = (1+x)^{-r\alpha} \sum_{k=0}^{\infty} \binom{-\alpha}{k}^r \left(\frac{x}{1+x} \right)^{rk},$$

the universal even-power complete-monotonicity conjecture is false. Already at $\alpha = 1$ and $r = 8$,

$$f_1^{[8]}(x) = \frac{1}{(1+x)^8 - x^8}$$

is not completely monotone on $(0, \infty)$.

Proof. For even $r = 2m$ and $\alpha = 1$, the source definition reduces to

$$f_1^{[2m]}(x) = \frac{1}{(1+x)^{2m} - x^{2m}}.$$

Let $D_m(x) = (1+x)^{2m} - x^{2m}$. Its roots are

$$\lambda_{k,m} = -\frac{1}{2} - \frac{i}{2} \cot \frac{\pi k}{2m}, \quad k = 1, \dots, 2m-1,$$

and the corresponding residues are

$$R_{k,m} = \frac{(-1)^{m+k}}{2m} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2}.$$

Thus the inverse-Laplace density of $1/D_m$, if it were nonnegative, would be

$$\rho_m(t) = \frac{e^{-t/2}}{2m} \left[2^{2m-2} + 2 \sum_{k=1}^{m-1} (-1)^{m+k} \left(2 \sin \frac{\pi k}{2m} \right)^{2m-2} \cos \left(\frac{t}{2} \cot \frac{\pi k}{2m} \right) \right].$$

For $m = 4$, write $h_4(t) = e^{t/2}\rho_4(t)$. The formula simplifies to

$$h_4(t) = 8 + 2 \cos \frac{t}{2} - \frac{(2 - \sqrt{2})^3}{4} \cos \left(\frac{1 + \sqrt{2}}{2} t \right) \\ - \frac{(2 + \sqrt{2})^3}{4} \cos \left(\frac{\sqrt{2} - 1}{2} t \right).$$

At $t = 10\pi$,

$$h_4(10\pi) = 6 + 10 \cos(5\pi\sqrt{2}).$$

Since $0 < 5\sqrt{2} - 7 < 1/4$,

$$\cos(5\pi\sqrt{2}) = -\cos(\pi(5\sqrt{2} - 7)) < -\frac{\sqrt{2}}{2} < -\frac{3}{5}.$$

Consequently $h_4(10\pi) < 0$, and so $\rho_4(10\pi) < 0$. By uniqueness of Laplace transforms of positive measures, $f_1^{[8]}$ cannot be completely monotone. $\square$

Theorem 12.14 (APP-0027: Ma–Weigert derivative regions form a chain). *Let*

$$\mathcal{F}_{1,n} = \left\{ x \mapsto \frac{p(\log x)}{x} : p \in \mathbb{R}[y]_n \right\},$$

and let $L = -d/dx$. If

$$D_k = \{f \in \mathcal{F}_{1,n} : L^k f(x) \geq 0 \text{ for all } x > 0\},$$

then

$$D_{k+1} \subseteq D_k \quad (k \geq 0, n \geq 0).$$

Proof. For $f(x) = p(\log x)/x$, induction gives

$$L^k f(x) = x^{-k-1} p_k(\log x)$$

for a polynomial p_k . Indeed, if the formula holds at level k , then

$$L^{k+1} f(x) = x^{-k-2} ((k+1)p_k(\log x) - p'_k(\log x)).$$

Thus $L^k f(x) \rightarrow 0$ as $x \rightarrow \infty$. If $f \in D_{k+1}$, put $g = L^k f$. Then $-g' = L^{k+1} f \geq 0$, and hence, for $R > x$,

$$g(x) - g(R) = \int_x^R L^{k+1} f(t) dt.$$

Letting $R \rightarrow \infty$ gives

$$L^k f(x) = \int_x^\infty L^{k+1} f(t) dt \geq 0.$$

Therefore $f \in D_k$. $\square$

Theorem 12.15 (APP-0028: divisor-polygamma parity correction). *For $n \in \mathbb{N}$, set*

$$f_n(x) = \sum_{km=n} [\psi^{(k)}(x)]^m, \quad x > 0.$$

Then f_n is completely monotone for every odd n . The asserted opposite parity law that $f_{2\ell}$ is never completely monotone is false:

$$f_2(x) = [\psi'(x)]^2 + \psi''(x)$$

is completely monotone on $(0, \infty)$.

Proof. The standard polygamma representation gives, for $k \geq 1$,

$$(-1)^{k+1}\psi^{(k)}(x) = \int_0^\infty \frac{t^k e^{-xt}}{1 - e^{-t}} dt.$$

Thus $\psi^{(k)}$ is a positive completely monotone function whenever k is odd. If n is odd and $km = n$, both k and m are odd, so each summand $[\psi^{(k)}]^m$ is a finite product of positive completely monotone functions. Finite sums preserve complete monotonicity.

For $n = 2$, the divisor pairs are $(1, 2)$ and $(2, 1)$, hence

$$f_2(x) = [\psi'(x)]^2 + \psi''(x).$$

Qi–Agarwal record the sharp theorem that

$$[\psi'(x)]^2 + \lambda\psi''(x)$$

is completely monotone on $(0, \infty)$ if and only if $\lambda \leq 1$. Taking $\lambda = 1$ proves that f_2 is completely monotone. This refutes the stated even-parity clause at $\ell = 1$. $\square$

Theorem 12.16 (APP-0029: Bulboacă–Zayed Gamma quotient one-critical-point theorem). *Let*

$$F(x) = \frac{\log \Gamma(x+1)}{\log(x^2+6) - \log(x+6)}$$

with the removable values filled in on $(-1, \infty)$. Then F' has a unique zero x_m on $(-1, \infty)$. Moreover

$$F'(x) < 0 \quad (-1 < x < x_m), \quad F'(x) > 0 \quad (x > x_m),$$

and numerically $x_m = 1.126207061051643 \dots$

Proof. The source proves $F' < 0$ on $(-1, 1)$. For $x > 1$, set

$$G(x) = \log \Gamma(x+1), \quad D(x) = \log \frac{x^2+6}{x+6},$$

and

$$N_0(x) = \psi(x+1)D(x) - G(x)D'(x).$$

Since $F' = N_0/D^2$, the sign of F' on $x > 1$ is the sign of N_0 . A direct right-tail estimate gives $N_0(x) > 0$ for $x \geq 8$: with

$$P(x) = x^2 + 12x - 6, \quad Q(x) = (x^2 + 6)(x + 6),$$

one has $D' = P/Q$; the bounds $\psi(x+1) > \log x$, $\log \Gamma(x+1) \leq x \log x$, and

$$D(x) \geq \frac{xP(x)}{Q(x)} \quad (x \geq 8)$$

imply $N_0(x) > 0$.

It remains to prove a single crossing on $(1, 8]$. Define

$$r(x) = \frac{\psi(x+1)}{D'(x)} = \psi(x+1) \frac{Q(x)}{P(x)}.$$

Then

$$P(x)^2 r'(x) = S(x),$$

where

$$S(x) = \psi'(x+1)Q(x)P(x) + \psi(x+1)\{Q'(x)P(x) - Q(x)P'(x)\}.$$

Using the elementary bounds, valid for $y \geq 2$,

$$\psi(y) > \log y - \frac{1}{y}, \quad \psi'(y) > \frac{1}{y} + \frac{1}{2y^2}, \quad \psi''(y) > -\frac{1}{y^2} - \frac{2}{y^3},$$

one obtains $S'(x) > 0$ on $[1, 8]$. Also

$$S(1) = 343 \left(\frac{\pi^2}{6} - 1 \right) - 539(1 - \gamma) < 0,$$

while $S(8) > 0$. Hence r' has exactly one zero on $[1, 8]$.

Now put $I(x) = D(x)r(x) - G(x)$. Then $N_0(x) = D'(x)I(x)$ and

$$I'(x) = D(x)r'(x).$$

Since $D(x) > 0$ for $x > 1$, I decreases once and then increases once. The removable endpoint has $\lim_{x \downarrow 1} I(x) = 0$, and the quadratic leading coefficient of N_0 at 1 is negative. Finally,

$$N_0(8) = (H_8 - \gamma) \log 5 - \frac{11}{70} \log(40320) > 0.$$

Thus N_0 , and hence F' , has exactly one zero on $(1, 8]$. Combining this with the source interval $(-1, 1)$ and the right-tail estimate proves the theorem. $\square$

Theorem 12.17 (APP-0030: Ramanujan Turan window has a negative answer). *Let*

$$I_R(x) = \int_0^\infty e^{-xt} \frac{dt}{t(\pi^2 + \log^2 t)}$$

and, for $n \geq 2$,

$$H_n(x; \alpha) = (I_R^{(n)}(x))^2 - \alpha I_R^{(n-1)}(x) I_R^{(n+1)}(x).$$

The proposed complete-monotonicity window

$$\frac{n-2}{n-1} < \alpha < \frac{n-1}{n}$$

is false. It already fails for $n = 2$.

Proof. Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t} dt.$$

Then

$$H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x).$$

Let $x = e^{-L}$. After the substitution $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2} dy.$$

Splitting at $|\log y| \leq L^{2/3}$ and expanding the denominator uniformly on the central region gives, for fixed $k = 1, 2, 3$,

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2}) \right).$$

Consequently

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} = \frac{1}{2} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2}) \right).$$

Since

$$\psi(1) = -\gamma, \quad \psi(2) = 1 - \gamma, \quad \psi(3) = \frac{3}{2} - \gamma,$$

the bracketed correction is $-1/L + O(L^{-2})$, and hence

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} = \frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose $\alpha_L = 1/2 - 1/(4L)$. For all sufficiently large L , $\alpha_L \in (0, 1/2)$ and

$$A_2(e^{-L})^2 - \alpha_L A_1(e^{-L}) A_3(e^{-L}) < 0.$$

Thus $H_2(e^{-L}; \alpha_L) < 0$. Complete monotonicity would imply nonnegativity at order zero, so the source's universal interval statement is false. $\square$

13 Late source applications

The following applications are already admitted in the Theory graph and are placed together because they are source-facing consequences of mechanisms developed earlier: Ramanujan moment-ratio asymptotics, Bessel zero partial fractions, and local endpoint obstructions.

Counterexample 13.1 (APP-0039: Ramanujan Turan-window complete-monotonicity interval is false). The Mishra–Swaminathan Ramanujan integral Turan-window complete-monotonicity statement has a negative answer: the proposed interval already fails for $n = 2$ and suitable $\alpha \in (0, 1/2)$. This resolves the source problem recorded as **APP-0039** [13].

Proof. Put

$$A_k(x) = (-1)^k I_R^{(k)}(x) = \int_0^\infty e^{-xt} \frac{t^{k-1}}{\pi^2 + \log^2 t} dt \quad (k \geq 1).$$

Then $H_2(x; \alpha) = A_2(x)^2 - \alpha A_1(x) A_3(x)$. Set $x = e^{-L}$. With $y = xt$,

$$A_k(e^{-L}) = e^{kL} \int_0^\infty \frac{e^{-y} y^{k-1}}{\pi^2 + (L + \log y)^2} dy.$$

For $k = 1, 2, 3$, splitting the integral into $|\log y| \leq L^{2/3}$ and its complement gives

$$A_k(e^{-L}) = e^{kL} L^{-2} \Gamma(k) \left(1 - \frac{2\psi(k)}{L} + O(L^{-2}) \right).$$

Therefore

$$\frac{A_2(e^{-L})^2}{A_1(e^{-L})A_3(e^{-L})} = \frac{\Gamma(2)^2}{\Gamma(1)\Gamma(3)} \left(1 + \frac{2(\psi(1) + \psi(3) - 2\psi(2))}{L} + O(L^{-2}) \right).$$

Using $\psi(1) = -\gamma$, $\psi(2) = 1 - \gamma$, and $\psi(3) = 3/2 - \gamma$, this ratio equals

$$\frac{1}{2} - \frac{1}{2L} + O(L^{-2}).$$

Choose $\alpha_L = 1/2 - 1/(4L)$. For all sufficiently large L , $\alpha_L \in (0, 1/2)$ and $H_2(e^{-L}; \alpha_L) < 0$. Complete monotonicity would imply nonnegativity, so the universal Turan-window statement is false. $\square$

Proposition 13.2 (Bessel W_ν zero representation). *For every $\nu > -1$, if $j_{\nu+1,n}$ are the positive zeros of $J_{\nu+1}$ and $W_\nu(z) = zI_\nu(z)/I_{\nu+1}(z)$, then*

$$G_\nu(s) := W_\nu(\sqrt{s}) = 2(\nu + 1) + 2 \sum_{n=1}^{\infty} \frac{s}{s + j_{\nu+1,n}^2},$$

and

$$G'_\nu(s) = 2 \sum_{n=1}^{\infty} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2}.$$

In particular G_ν is a Bernstein function.

Proof. Put $\mu = \nu + 1 > 0$. The canonical product for I_μ gives

$$I_\mu(z) = \frac{(z/2)^\mu}{\Gamma(\mu + 1)} \prod_{n=1}^{\infty} \left(1 + \frac{z^2}{j_{\mu,n}^2}\right).$$

Hence

$$z \frac{I'_\mu(z)}{I_\mu(z)} = \mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

Using $I'_\mu(z) = I_{\mu-1}(z) - \mu I_\mu(z)/z$, we obtain

$$W_\nu(z) = z \frac{I_{\mu-1}(z)}{I_\mu(z)} = 2\mu + 2 \sum_{n=1}^{\infty} \frac{z^2}{z^2 + j_{\mu,n}^2}.$$

Set $s = z^2$. Since $j_{\nu+1,n} \sim \pi n$, the differentiated sums converge locally uniformly, and

$$(-1)^k \frac{d^k}{ds^k} \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^2} = (k+1)! \frac{j_{\nu+1,n}^2}{(s + j_{\nu+1,n}^2)^{k+2}} \geq 0.$$

Thus G'_ν is completely monotone, and G_ν is Bernstein. $\square$

Corollary 13.3 (APP-0040: Yang-Tian Bessel- W power-Bernstein conjecture is true). *For $\tau \in (0, 1/2]$ and $\nu > -1$, $x \mapsto W_\nu(x^\tau)$ is a Bernstein function on $(0, \infty)$. This resolves the source problem recorded as **APP-0040** [34].*

Proof. By Proposition 13.2, G_ν is Bernstein. If $0 < \tau \leq 1/2$, then $x^{2\tau}$ is Bernstein. The Bernstein composition closure used in Theorem 12.6 gives

$$W_\nu(x^\tau) = G_\nu(x^{2\tau}),$$

which is Bernstein. $\square$

Counterexample 13.4 (APP-0041: Qi h_λ degree-four complete-monotonicity conjecture is false).
Let

$$\Psi(x) = [\psi'(x)]^2 + \psi''(x), \quad h_\lambda(x) = \Psi(x) - \frac{x^2 + \lambda x + 12}{12x^4(x+1)^2}.$$

The conjecture that $\deg_x^{\text{cm}} h_\lambda = \deg_x^{\text{cm}}(-h_\mu) = 4$ exactly for $\lambda \leq 0$ and $\mu \geq 4$ is false. This resolves the source problem recorded as **APP-0041** [19].

Proof. At $x = 0^+$,

$$\psi'(x) = \frac{1}{x^2} + \zeta(2) - 2\zeta(3)x + 3\zeta(4)x^2 + O(x^3), \quad \psi''(x) = -\frac{2}{x^3} - 2\zeta(3) + 6\zeta(4)x + O(x^2).$$

Thus

$$\Psi(x) = \frac{1}{x^4} - \frac{2}{x^3} + \frac{\pi^2}{3x^2} - \frac{4\zeta(3)}{x} + O(1).$$

Also

$$\frac{x^2 + \lambda x + 12}{12x^4(1+x)^2} = \frac{1}{x^4} + \left(\frac{\lambda}{12} - 2\right) \frac{1}{x^3} + \left(\frac{37}{12} - \frac{\lambda}{6}\right) \frac{1}{x^2} + O(x^{-1}).$$

Hence

$$h_\lambda(x) = -\frac{\lambda}{12x^3} + \left(\frac{\lambda}{6} + \frac{\pi^2}{3} - \frac{37}{12}\right) \frac{1}{x^2} + O(x^{-1}).$$

If $\lambda < 0$, then $\frac{d}{dx}[x^4 h_\lambda(x)] = -\lambda/12 + O(x) > 0$ for small $x > 0$, so $x^4 h_\lambda$ is not completely monotone. If $\lambda = 0$, then

$$x^4 h_0(x) = \left(\frac{\pi^2}{3} - \frac{37}{12}\right) x^2 + O(x^3),$$

and $4\pi^2 > 37$ again gives a positive first derivative near zero. Similarly, for $\mu \geq 4$,

$$\frac{d}{dx}[x^4(-h_\mu(x))] = \frac{\mu}{12} + O(x) > 0$$

near zero. The degree-four transforms required by the conjecture therefore fail on the conjectured source ranges. $\square$

14 Bessel endpoint obstruction

The next group belongs to the same endpoint-obstruction mechanism used throughout the paper: a universal positivity or concavity statement is reduced to a one-point Riccati certificate, and exact rational bounds then decide the sign. The same certificate resolves the square-root log-concavity problem, the Riccati form of that problem, a related quadratic ratio bound, and the proposed three-regime certificate route.

Counterexample 14.1 (APP-0042: Modified Bessel square-root log-concavity fails). The assertion that, for every $\nu \geq 0$, the function

$$u \longmapsto \sqrt{u} I_\nu(u)$$

is strictly log-concave on $(0, \infty)$ is false. More precisely, at $(\nu, u) = (0, 10)$ the logarithmic second derivative is positive. This resolves the source problem recorded as **APP-0042** [35].

Proof. Let

$$g_\nu(u) = \log(\sqrt{u} I_\nu(u)), \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)}.$$

Then

$$g''_\nu(u) = r'_\nu(u) - \frac{1}{2u^2}.$$

The modified Bessel equation

$$u^2 I''_\nu(u) + u I'_\nu(u) - (u^2 + \nu^2) I_\nu(u) = 0$$

gives

$$\frac{I''_\nu(u)}{I_\nu(u)} = 1 + \frac{\nu^2}{u^2} - \frac{r_\nu(u)}{u},$$

and therefore

$$g''_\nu(u) = 1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2.$$

For $\nu = 0$, $r_0(u) = I_1(u)/I_0(u)$. At $u = 10$,

$$g''_0(10) = \frac{199}{200} - \frac{r_0(10)}{10} - r_0(10)^2.$$

It is enough to prove

$$r_0(10) < \frac{9487}{10000},$$

because

$$\frac{199}{200} - \frac{1}{10} \frac{9487}{10000} - \left(\frac{9487}{10000} \right)^2 = \frac{9831}{100000000} > 0.$$

The modified Bessel series are

$$I_0(10) = \sum_{k=0}^{\infty} \frac{25^k}{(k!)^2}, \quad I_1(10) = \sum_{k=0}^{\infty} \frac{5 \cdot 25^k}{k!(k+1)!}.$$

Using the first 21 terms as a lower bound for $I_0(10)$, and bounding the I_1 -tail after $k = 20$ by a geometric tail with ratio

$$q = \frac{25}{22 \cdot 23} < 1,$$

the audited rational certificate gives an upper bound $U_1 > I_1(10)$ satisfying

$$10000 U_1 < 9487 \sum_{k=0}^{20} \frac{25^k}{(k!)^2}.$$

Hence $I_1(10)/I_0(10) < 9487/10000$, so $g''_0(10) > 0$. The claimed universal strict log-concavity is therefore refuted. $\square$

Corollary 14.2 (APP-0043: Riccati log-concavity inequality for I_ν is false). *The inequality*

$$1 + \frac{\nu^2 - \frac{1}{2}}{u^2} - \frac{r_\nu(u)}{u} - r_\nu(u)^2 < 0, \quad r_\nu(u) = \frac{I'_\nu(u)}{I_\nu(u)},$$

*does not hold for every $\nu \geq 0$ and $u > 0$. This resolves the source problem recorded as **APP-0043** [35].*

Proof. The proof of Counterexample 14.1 gives

$$1 - \frac{1}{2 \cdot 10^2} - \frac{r_0(10)}{10} - r_0(10)^2 = g_0''(10) > 0.$$

This is the negation of the claimed strict Riccati inequality at $(\nu, u) = (0, 10)$. $\square$

Counterexample 14.3 (APP-0044: Bessel ratio quadratic lower bound is false). The claimed inequality

$$q_\nu(u)^2 + \frac{(2\nu + 1)q_\nu(u)}{u} + \frac{\nu + \frac{1}{2}}{u^2} > 1, \quad q_\nu(u) = \frac{I_{\nu+1}(u)}{I_\nu(u)},$$

is false for all $\nu \geq 0$ and $u > 0$. This resolves the source problem recorded as **APP-0044** [35].

Proof. At $\nu = 0$ and $u = 10$, the same rational certificate used above gives

$$q_0(10) = \frac{I_1(10)}{I_0(10)} < \frac{9487}{10000}.$$

Therefore

$$q_0(10)^2 + \frac{q_0(10)}{10} + \frac{1}{200} < \left(\frac{9487}{10000}\right)^2 + \frac{1}{10} \frac{9487}{10000} + \frac{1}{200} = 1 - \frac{9831}{100000000} < 1.$$

This contradicts the proposed strict lower bound. $\square$

Corollary 14.4 (APP-0045: Three-regime Bessel log-concavity certificate route is refuted). *No valid three-regime certificate can prove the universal Riccati log-concavity inequality for I_ν , because the universal inequality itself is false. This resolves the source problem recorded as **APP-0045** [35].*

Proof. Corollary 14.2 gives an explicit point $(\nu, u) = (0, 10)$ where the target inequality fails. A certificate proving the inequality for all $\nu \geq 0$ and $u > 0$ would imply the false universal statement. Hence the proposed certificate route cannot exist. $\square$

15 Conclusion

The theory developed here is organized around a single principle: positivity should be transported only through mechanisms that preserve the relevant analytic structure. Zeta-law compression turns Dirichlet sums into probabilistic identities, special-function curvature is tested through Laplace kernels, and application-level claims are resolved by moving between Stieltjes, Bernstein, determinant, ratio, and endpoint-obstruction forms.

This perspective gives the paper its hierarchy. Definitions fix the common objects, lemmas and propositions isolate reusable transformations, theorems state the main transport mechanisms, corollaries and counterexamples record their consequences, and the application ledger points each source problem to the result that resolves it. The result is a connected theory of positivity transport rather than a list of unrelated solved examples.

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